

Differential Calculus

Unit I — Lecture Notes

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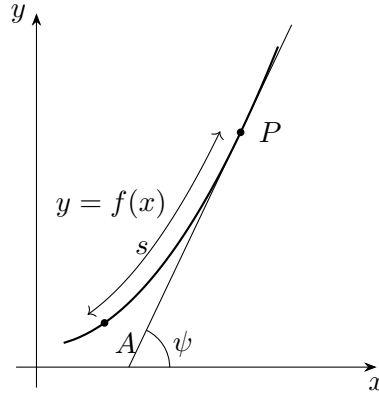
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1 Curvature

Definition 1.1 (Curvature). The *curvature* of a curve $y = f(x)$ gives information about the rate of change of the curve at its points, that is, the rate at which the curve changes (bends) at each and every point of the curve.

A curve's curvature can be measured as the rate of change of the tangent angle ψ at a point P with respect to the arc length s .



From the figure, the curvature of the curve at the point P is $\frac{d\psi}{ds}$. The curvature of a curve is always denoted by κ (kappa):

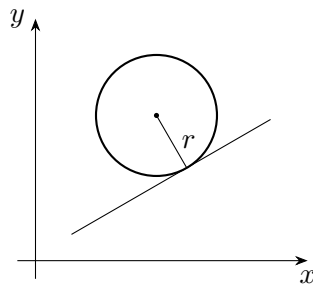
$$\kappa = \text{curvature at } P = \frac{d\psi}{ds}.$$

Example (circle). Consider a circle of radius r . For this circle the arc length is $s = r\psi$, so

$$\psi = \frac{s}{r}, \quad \text{Curvature} = \frac{d\psi}{ds} = \frac{1}{r}.$$

Hence, for any circle, the curvature at any point of the circle is the reciprocal of its radius. That is,

$$\text{Radius of circle} = \frac{1}{\text{Curvature of the circle}}.$$



2 Radius of Curvature

Definition 2.1 (Radius of curvature). The *radius of curvature* of any curve is defined as the reciprocal of its curvature. It is denoted by ρ :

$$\rho = \frac{1}{\text{curvature of the curve}}, \quad \text{i.e.} \quad \rho = \frac{1}{\kappa}.$$

2.1 Formulae for the radius of curvature

1. Cartesian form $y = f(x)$:

$$\rho = \frac{(1 + y_1^2)^{3/2}}{y_2}, \quad y_1 = \left. \frac{dy}{dx} \right|_{(x,y)}, \quad y_2 = \left. \frac{d^2y}{dx^2} \right|_{(x,y)}.$$

2. Parametric form $x = x(t), y = y(t)$:

$$\rho = \frac{(\dot{x}^2 + \dot{y}^2)^{3/2}}{\dot{x}\ddot{y} - \dot{y}\ddot{x}}, \quad \dot{x} = \frac{dx}{dt}, \quad \ddot{x} = \frac{d^2x}{dt^2}, \quad \dot{y} = \frac{dy}{dt}, \quad \ddot{y} = \frac{d^2y}{dt^2}.$$

Note. • The radius of curvature of a curve is always taken as a *positive* quantity.
 • When finding the radius of curvature in Cartesian form, if $\frac{dy}{dx}$ at a point (x, y) is infinite or undefined, then y is treated as the independent variable and we use

$$\rho = \frac{(1 + x_1^2)^{3/2}}{x_2}, \quad x_1 = \left. \frac{dx}{dy} \right|_{(x,y)}, \quad x_2 = \left. \frac{d^2x}{dy^2} \right|_{(x,y)}.$$

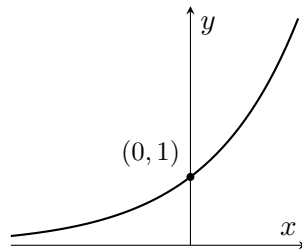
2.2 Problems in Cartesian form

Problem 2.1. Find the radius of curvature of the curve $y = e^x$ at $(0, 1)$.

Solution. $y = e^x \Rightarrow \frac{dy}{dx} = e^x$, so $y_1 = e^0 = 1$. Also $\frac{d^2y}{dx^2} = e^x$, so $y_2 = e^0 = 1$.

$$\rho = \frac{(1 + y_1^2)^{3/2}}{y_2} = \frac{(1 + 1)^{3/2}}{1} = 2^{3/2} = 2\sqrt{2} \text{ units.}$$

Hence the curvature of the given curve at $(0, 1)$ is $\kappa = \frac{1}{\rho} = \frac{1}{2\sqrt{2}}$ units.



Problem 2.2. What is the radius of curvature of the circle $x^2 + y^2 = 25$?

Solution. We know that for a circle, the radius of curvature is its radius. Hence $\rho = 5$.

Problem 2.3. Find the radius of curvature of $y = c \cosh(x/c)$, where c is a constant.

Solution.

$$\frac{dy}{dx} = c \sinh(x/c) \cdot \frac{1}{c} \Rightarrow y_1 = \sinh(x/c), \quad \frac{d^2y}{dx^2} = \cosh(x/c) \cdot \frac{1}{c} \Rightarrow y_2 = \frac{1}{c} \cosh(x/c).$$

Using $1 + \sinh^2 x = \cosh^2 x$,

$$\rho = \frac{[1 + \sinh^2(x/c)]^{3/2}}{\frac{1}{c} \cosh(x/c)} = \frac{c (\cosh^2(x/c))^{3/2}}{\cosh(x/c)} = \frac{c \cosh^3(x/c)}{\cosh(x/c)} = c \cosh^2(x/c).$$

From the curve, $\cosh(x/c) = \frac{y}{c}$. Therefore

$$\rho = c \left(\frac{y}{c} \right)^2 = c \cdot \frac{y^2}{c^2} = \frac{y^2}{c}.$$

Problem 2.4. Find the radius of curvature of the curve $\sqrt{x} + \sqrt{y} = 1$ at $(\frac{1}{4}, \frac{1}{4})$.

Solution. Differentiating $\sqrt{x} + \sqrt{y} = 1$ with respect to x :

$$\frac{1}{2}x^{-1/2} + \frac{1}{2}y^{-1/2} \frac{dy}{dx} = 0 \Rightarrow \frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}} \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{\sqrt{y}}{\sqrt{x}}.$$

At $(\frac{1}{4}, \frac{1}{4})$: $y_1 = -\frac{\sqrt{1/4}}{\sqrt{1/4}} = -1$.

By the quotient rule $\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{vu' - uv'}{v^2}$,

$$\frac{d^2y}{dx^2} = - \left[\frac{\sqrt{x} \cdot \frac{1}{2}y^{-1/2} \frac{dy}{dx} - \sqrt{y} \cdot \frac{1}{2}x^{-1/2}}{(\sqrt{x})^2} \right] = - \left[\frac{\sqrt{x} \cdot \frac{1}{2\sqrt{y}} \frac{dy}{dx} - \frac{\sqrt{y}}{2\sqrt{x}}}{x} \right].$$

At $(\frac{1}{4}, \frac{1}{4})$ with $\frac{dy}{dx} = -1$:

$$y_2 = -4 \left[\frac{1}{2} \cdot \frac{1}{2 \cdot \frac{1}{2}} (-1) - \frac{1}{2} \right] = -4 \left[-\frac{1}{2} - \frac{1}{2} \right] = -4(-1) = 4.$$

Hence

$$\rho = \frac{(1 + y_1^2)^{3/2}}{y_2} = \frac{(1 + (-1)^2)^{3/2}}{4} = \frac{2^{3/2}}{4} = \frac{2\sqrt{2}}{4} = \frac{1}{\sqrt{2}} \text{ units.}$$

Problem 2.5. Find the radius of curvature of the curve $x^3 + y^3 = 3xy$ at $(\frac{1}{2}, \frac{1}{2})$.

Solution. Differentiating with respect to x (product rule $\frac{d}{dx}(uv) = uv' + vu'$):

$$3x^2 + 3y^2 \frac{dy}{dx} = 3 \left(x \frac{dy}{dx} + y \right) \Rightarrow \frac{dy}{dx} (y^2 - x) = y - x^2 \Rightarrow \frac{dy}{dx} = \frac{y - x^2}{y^2 - x} = \frac{x^2 - y}{x - y^2}.$$

At $(\frac{1}{2}, \frac{1}{2})$:

$$y_1 = \frac{(\frac{1}{2})^2 - \frac{1}{2}}{\frac{1}{2} - (\frac{1}{2})^2} = \frac{\frac{1}{4} - \frac{1}{2}}{\frac{1}{2} - \frac{1}{4}} = \frac{-1/4}{1/4} = -1.$$

Differentiating again,

$$\frac{d^2y}{dx^2} = \frac{(x - y^2) \left(2x - \frac{dy}{dx} \right) - (x^2 - y) \left(1 - 2y \frac{dy}{dx} \right)}{(x - y^2)^2}.$$

At $(\frac{1}{2}, \frac{1}{2})$:

$$y_2 = \frac{(\frac{1}{2} - \frac{1}{4}) (2 \cdot \frac{1}{2} - (-1)) - (\frac{1}{4} - \frac{1}{2}) (1 - 2 \cdot \frac{1}{2}(-1))}{(\frac{1}{2} - \frac{1}{4})^2} = \frac{(\frac{1}{4}) (2) - (-\frac{1}{4}) (2)}{(\frac{1}{4})^2} = \frac{1}{1/16} = 16.$$

Hence

$$\rho = \frac{(1 + y_1^2)^{3/2}}{y_2} = \frac{(1 + 1)^{3/2}}{16} = \frac{2\sqrt{2}}{16} = \frac{\sqrt{2}}{8}.$$

Problem 2.6. If ρ is the radius of curvature at any point (x, y) on the curve $y = \frac{ax}{a+x}$, show that

$$\left(\frac{2\rho}{a} \right)^{2/3} = \left(\frac{x}{y} \right)^2 + \left(\frac{y}{x} \right)^2.$$

Solution.

$$\frac{dy}{dx} = \frac{(a+x)(a) - (ax)(0+1)}{(a+x)^2} = \frac{a^2 + ax - ax}{(a+x)^2} \Rightarrow y_1 = \frac{a^2}{(a+x)^2}.$$

$$\frac{d^2y}{dx^2} = \frac{(a+x)^2(0) - a^2 \cdot 2(a+x)(0+1)}{(a+x)^4} = \frac{-2a^2(a+x)}{(a+x)^4} \Rightarrow y_2 = \frac{-2a^2}{(a+x)^3}.$$

Since ρ is positive, we use $|y_2|$:

$$\rho = \frac{\left(1 + \frac{a^4}{(a+x)^4} \right)^{3/2}}{\left| \frac{-2a^2}{(a+x)^3} \right|} = \frac{(a+x)^3}{2a^2} \left(\frac{(a+x)^4 + a^4}{(a+x)^4} \right)^{3/2}.$$

Multiplying both sides by $\frac{2}{a}$,

$$\frac{2\rho}{a} = \frac{(a+x)^3}{a^3} \left[\frac{(a+x)^4 + a^4}{(a+x)^4} \right]^{3/2}.$$

Raising both sides to the power $\frac{2}{3}$,

$$\left(\frac{2\rho}{a} \right)^{2/3} = \left(\frac{a+x}{a} \right)^{3 \cdot \frac{2}{3}} \left(\frac{(a+x)^4 + a^4}{(a+x)^4} \right)^{\frac{3}{2} \cdot \frac{2}{3}} = \frac{(a+x)^2}{a^2} \cdot \frac{(a+x)^4 + a^4}{(a+x)^4}$$

$$\begin{aligned}
&= \frac{(a+x)^4 + a^4}{a^2(a+x)^2} = \frac{(a+x)^4}{a^2(a+x)^2} + \frac{a^4}{a^2(a+x)^2} = \frac{(a+x)^2}{a^2} + \frac{a^2}{(a+x)^2} \\
&= \left(\frac{a+x}{a}\right)^2 + \left(\frac{a}{a+x}\right)^2.
\end{aligned}$$

We know $y = \frac{ax}{a+x}$, so $\frac{y}{x} = \frac{a}{a+x}$ and $\frac{x}{y} = \frac{a+x}{a}$. Therefore

$$\left(\frac{2\rho}{a}\right)^{2/3} = \left(\frac{x}{y}\right)^2 + \left(\frac{y}{x}\right)^2. \quad \text{Hence proved.}$$

2.3 Problems in parametric form

Problem 2.7. Find the radius of curvature of the astroid $x^{2/3} + y^{2/3} = a^{2/3}$ using its parametric form.

Solution. The parametric form of the astroid is $x = a \cos^3 \theta$, $y = a \sin^3 \theta$.

$$\begin{aligned}
\dot{x} &= \frac{dx}{d\theta} = -3a \cos^2 \theta \sin \theta, \\
\ddot{x} &= -3a (\cos^2 \theta \cos \theta + \sin \theta \cdot 2 \cos \theta (-\sin \theta)) = -3a (\cos^3 \theta - 2 \cos \theta \sin^2 \theta), \\
\dot{y} &= \frac{dy}{d\theta} = 3a \sin^2 \theta \cos \theta, \\
\ddot{y} &= 3a (\sin^2 \theta (-\sin \theta) + \cos \theta \cdot 2 \sin \theta \cos \theta) = 3a (2 \cos^2 \theta \sin \theta - \sin^3 \theta).
\end{aligned}$$

Now

$$\begin{aligned}
\dot{x}^2 + \dot{y}^2 &= 9a^2 \cos^4 \theta \sin^2 \theta + 9a^2 \sin^4 \theta \cos^2 \theta \\
&= 9a^2 \sin^2 \theta \cos^2 \theta (\cos^2 \theta + \sin^2 \theta) = 9a^2 \sin^2 \theta \cos^2 \theta, \\
(\dot{x}^2 + \dot{y}^2)^{3/2} &= ((3a \sin \theta \cos \theta)^2)^{3/2} = (3a \sin \theta \cos \theta)^3. \\
\dot{x}\ddot{y} - \dot{y}\ddot{x} &= (-3a \cos^2 \theta \sin \theta) 3a (2 \cos^2 \theta \sin \theta - \sin^3 \theta) \\
&\quad - (3a \sin^2 \theta \cos \theta) (-3a) (\cos^3 \theta - 2 \cos \theta \sin^2 \theta) \\
&= -18a^2 \cos^4 \theta \sin^2 \theta + 9a^2 \cos^2 \theta \sin^4 \theta \\
&\quad + 9a^2 \cos^4 \theta \sin^2 \theta - 18a^2 \sin^4 \theta \cos^2 \theta \\
&= -9a^2 \cos^4 \theta \sin^2 \theta - 9a^2 \sin^4 \theta \cos^2 \theta \\
&= -9a^2 \sin^2 \theta \cos^2 \theta (\sin^2 \theta + \cos^2 \theta) \\
&= -9a^2 \sin^2 \theta \cos^2 \theta.
\end{aligned}$$

Hence

$$\rho = \frac{(3a \sin \theta \cos \theta)^3}{-(3a \sin \theta \cos \theta)^2} = |-3a \sin \theta \cos \theta| \Rightarrow \rho = 3a \sin \theta \cos \theta \quad (\rho \text{ is always positive}).$$

Problem 2.8. Find the radius of curvature of the circle $x^2 + y^2 = a^2$.

Solution. The radius of curvature of a circle equals its radius, so $\rho = a$.

Problem 2.9. Find the radius of curvature of the cycloid $x = a(\theta - \sin \theta)$, $y = a(1 - \cos \theta)$.

Solution.

$$\begin{aligned}\dot{x} &= a(1 - \cos \theta), & \ddot{x} &= a(0 - (-\sin \theta)) = a \sin \theta, \\ \dot{y} &= a(0 - (-\sin \theta)) = a \sin \theta, & \ddot{y} &= a \cos \theta.\end{aligned}$$

$$\begin{aligned}\dot{x}^2 + \dot{y}^2 &= a^2(1 - \cos \theta)^2 + a^2 \sin^2 \theta = a^2(1 + \cos^2 \theta - 2 \cos \theta) + a^2 \sin^2 \theta \\ &= a^2 + a^2 - 2a^2 \cos \theta = 2a^2(1 - \cos \theta) = 2a^2 \left(2 \sin^2 \frac{\theta}{2}\right) = \left(2a \sin \frac{\theta}{2}\right)^2,\end{aligned}$$

$$\text{so } (\dot{x}^2 + \dot{y}^2)^{3/2} = \left(2a \sin \frac{\theta}{2}\right)^{2 \cdot \frac{3}{2}} = 8a^3 \sin^3 \frac{\theta}{2}.$$

$$\text{Using } \cos \theta = 1 - 2 \sin^2 \frac{\theta}{2},$$

$$\begin{aligned}\dot{x}\ddot{y} - \dot{y}\ddot{x} &= a(1 - \cos \theta) a \cos \theta - a \sin \theta \cdot a \sin \theta = a^2 \cos \theta - a^2 \cos^2 \theta - a^2 \sin^2 \theta \\ &= a^2 \cos \theta - a^2 = -a^2(1 - \cos \theta) = -a^2 \left(2 \sin^2 \frac{\theta}{2}\right) = -2a^2 \sin^2 \frac{\theta}{2}.\end{aligned}$$

Hence

$$\rho = \frac{8a^3 \sin^3 \frac{\theta}{2}}{-2a^2 \sin^2 \frac{\theta}{2}} = \left| -4a \sin \frac{\theta}{2} \right| \Rightarrow \rho = 4a \sin \frac{\theta}{2} \quad (\rho \text{ is positive}).$$

Problem 2.10. Find the radius of curvature of the parabola given in parametric form $x = at^2$, $y = 2at$.

Solution.

$$\dot{x} = 2at, \quad \ddot{x} = 2a, \quad \dot{y} = 2a, \quad \ddot{y} = 0.$$

$$\dot{x}^2 + \dot{y}^2 = (2at)^2 + (2a)^2 = 4a^2 t^2 + 4a^2 = 4a^2(1 + t^2),$$

$$(\dot{x}^2 + \dot{y}^2)^{3/2} = (2a)^{2 \cdot \frac{3}{2}}(1 + t^2)^{3/2} = 8a^3(1 + t^2)^{3/2}.$$

$$\dot{x}\ddot{y} - \dot{y}\ddot{x} = (2at)(0) - (2a)(2a) = -4a^2.$$

$$\rho = \frac{8a^3(1 + t^2)^{3/2}}{-4a^2} = \left| -2a(1 + t^2)^{3/2} \right| \Rightarrow \rho = 2a(1 + t^2)^{3/2} \quad (\text{since } \rho \text{ is positive}).$$

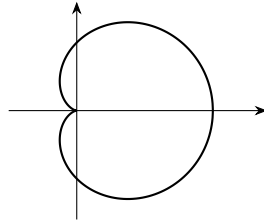
2.4 Radius of curvature in polar form

When the equation of the curve is in polar form $r = f(\theta)$, the radius of curvature is given by

$$\rho = \frac{(r^2 + r_1^2)^{3/2}}{r^2 - r r_2 + 2r_1^2}, \quad r_1 = \frac{dr}{d\theta}, \quad r_2 = \frac{d^2r}{d\theta^2}.$$

Problem 2.11. Find the radius of curvature of the cardioid $r = a(1 + \cos \theta)$ at $\theta = \frac{\pi}{2}$.

Solution.



$r = a(1 + \cos \theta)$, so $r_1 = \frac{dr}{d\theta} = a(0 - \sin \theta) = -a \sin \theta$. At $\theta = \frac{\pi}{2}$ ($\sin 90^\circ = 1$): $r_1 = -a$.
 $r_2 = -a \cos \theta$. At $\theta = \frac{\pi}{2}$ ($\cos 90^\circ = 0$): $r_2 = 0$. Also $r = a(1 + 0) = a$.

$$\rho = \frac{(a^2 + (-a)^2)^{3/2}}{a^2 - a \cdot 0 + 2(-a)^2} = \frac{(2a^2)^{3/2}}{3a^2} = \frac{2^{3/2}a^3}{3a^2} \Rightarrow \rho = \frac{2\sqrt{2}a}{3}.$$

Problem 2.12. Show that at the points of intersection of the curves $r = a\theta$ and $r = \frac{a}{\theta}$, their curvatures are in the ratio 3 : 1.

Solution. Points of intersection: $a\theta = \frac{a}{\theta} \Rightarrow \theta^2 = 1 \Rightarrow \theta = \pm 1$.

Curve $r = a\theta$: $r_1 = a$, $r_2 = 0$.

$$\rho = \frac{(a^2\theta^2 + a^2)^{3/2}}{a^2\theta^2 - a\theta \cdot 0 + 2a^2} = \frac{(a^2\theta^2 + a^2)^{3/2}}{a^2\theta^2 + 2a^2}.$$

At $\theta = \pm 1$:

$$\rho_1 = \frac{(a^2 + a^2)^{3/2}}{a^2 + 2a^2} = \frac{(2a^2)^{3/2}}{3a^2} = \frac{2^{3/2}a^3}{3a^2} = \frac{2\sqrt{2}a}{3}.$$

Curve $r = \frac{a}{\theta} = a\theta^{-1}$: $r_1 = -a\theta^{-2} = -\frac{a}{\theta^2}$, $r_2 = 2a\theta^{-3} = \frac{2a}{\theta^3}$.

$$\rho = \frac{\left(\left(\frac{a}{\theta}\right)^2 + \left(-\frac{a}{\theta^2}\right)^2\right)^{3/2}}{\left(\frac{a}{\theta}\right)^2 - \left(\frac{a}{\theta}\right)\left(\frac{2a}{\theta^3}\right) + 2\left(-\frac{a}{\theta^2}\right)^2}.$$

At $\theta = \pm 1$:

$$\rho_2 = \frac{(a^2 + a^2)^{3/2}}{a^2 - 2a^2 + 2a^2} = \frac{(2a^2)^{3/2}}{a^2} = \frac{2^{3/2}a^3}{a^2} = 2\sqrt{2}a.$$

Thus $\rho_1 : \rho_2 = \frac{2\sqrt{2}a}{3} : 2\sqrt{2}a = \frac{1}{3} : 1 = 1 : 3$. Since $\kappa = \frac{1}{\rho}$,

$$\kappa_1 : \kappa_2 = \frac{1}{\rho_1} : \frac{1}{\rho_2} = \frac{3}{2\sqrt{2}a} : \frac{1}{2\sqrt{2}a} = 3 : 1.$$

Therefore the curvatures of the given curves at their points of intersection are in the ratio 3 : 1.

Problem 2.13. Show that the radius of curvature of the curve $r^n = a^n \sin n\theta$ at any point

$$(r, \theta) \text{ is } \frac{a^n}{(n+1)r^{n-1}}.$$

Solution. Taking logarithms of $r^n = a^n \sin n\theta$:

$$n \log r = n \log a + \log \sin n\theta.$$

Differentiating with respect to θ :

$$n \cdot \frac{1}{r} \frac{dr}{d\theta} = 0 + \frac{1}{\sin n\theta} \cdot \cos n\theta \cdot n \Rightarrow \frac{r_1}{r} = \cot n\theta \Rightarrow r_1 = r \cot n\theta.$$

Differentiating again with respect to θ (product rule),

$$r_2 = r_1 \cot n\theta + r (-\operatorname{cosec}^2 n\theta) n = r \cot^2 n\theta - nr \operatorname{cosec}^2 n\theta \quad (\text{using } r_1 = r \cot n\theta).$$

Numerator. Using $\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$,

$$r^2 + r_1^2 = r^2 + r^2 \cot^2 n\theta = r^2(1 + \cot^2 n\theta) = r^2 \operatorname{cosec}^2 n\theta = (r \operatorname{cosec} n\theta)^2,$$

$$N_r = (r^2 + r_1^2)^{3/2} = (r \operatorname{cosec} n\theta)^{2 \cdot \frac{3}{2}} = r^3 \operatorname{cosec}^3 n\theta.$$

Denominator.

$$\begin{aligned} D_r &= r^2 - rr_2 + 2r_1^2 = r^2 - r(r \cot^2 n\theta - nr \operatorname{cosec}^2 n\theta) + 2(r \cot n\theta)^2 \\ &= r^2 - r^2 \cot^2 n\theta + nr^2 \operatorname{cosec}^2 n\theta + 2r^2 \cot^2 n\theta \\ &= r^2 + r^2 (\cot^2 n\theta + n \operatorname{cosec}^2 n\theta) \\ &= r^2 + r^2 (-1 + \operatorname{cosec}^2 n\theta + n \operatorname{cosec}^2 n\theta) \\ &= r^2 - r^2 + r^2 \operatorname{cosec}^2 n\theta (1 + n) = r^2 \operatorname{cosec}^2 n\theta (n + 1). \end{aligned}$$

Hence

$$\rho = \frac{N_r}{D_r} = \frac{r^3 \operatorname{cosec}^3 n\theta}{r^2 \operatorname{cosec}^2 n\theta (n + 1)} = \frac{r \operatorname{cosec} n\theta}{n + 1} = \frac{r}{n + 1} \cdot \frac{1}{\sin n\theta}.$$

Since $r^n = a^n \sin n\theta$, we have $\sin n\theta = \frac{r^n}{a^n}$, and so

$$\rho = \frac{r}{n + 1} \cdot \frac{a^n}{r^n} \Rightarrow \rho = \frac{a^n}{(n + 1)r^{n-1}}. \quad \text{Hence proved.}$$

Problem 2.14. Find the radius of curvature of the curve $r^2 \cos 2\theta = a^2$.

Solution. $r^2 = a^2 \sec 2\theta$. Taking logarithms, $2 \log r = 2 \log a + \log \sec 2\theta$. Differentiating with respect to θ :

$$2 \cdot \frac{1}{r} \frac{dr}{d\theta} = 0 + \frac{1}{\sec 2\theta} \cdot \sec 2\theta \tan 2\theta \cdot 2 \Rightarrow \frac{2}{r} r_1 = 2 \tan 2\theta \Rightarrow r_1 = r \tan 2\theta.$$

Differentiating again (product rule) and using $r_1 = r \tan 2\theta$:

$$r_2 = r (\sec^2 2\theta \cdot 2) + \tan 2\theta \cdot r_1 = 2r \sec^2 2\theta + r \tan^2 2\theta.$$

Numerator. Using $\sec^2 \theta - \tan^2 \theta = 1$,

$$r^2 + r_1^2 = r^2 + r^2 \tan^2 2\theta = r^2(1 + \tan^2 2\theta) = r^2 \sec^2 2\theta, \quad N_r = (r \sec 2\theta)^{2 \cdot \frac{3}{2}} = r^3 \sec^3 2\theta.$$

Denominator.

$$\begin{aligned}
 D_r &= r^2 - rr_2 + 2r_1^2 = r^2 - r(2r \sec^2 2\theta + r \tan^2 2\theta) + 2(r \tan 2\theta)^2 \\
 &= r^2 - 2r^2 \sec^2 2\theta - r^2 \tan^2 2\theta + 2r^2 \tan^2 2\theta = r^2 - 2r^2 \sec^2 2\theta + r^2 \tan^2 2\theta \\
 &= r^2 + r^2 (\tan^2 2\theta - 2 \sec^2 2\theta) = r^2 + r^2 (\sec^2 2\theta - 1 - 2 \sec^2 2\theta) \\
 &= r^2 + r^2 (-1 - \sec^2 2\theta) = r^2 - r^2 - r^2 \sec^2 2\theta = -r^2 \sec^2 2\theta.
 \end{aligned}$$

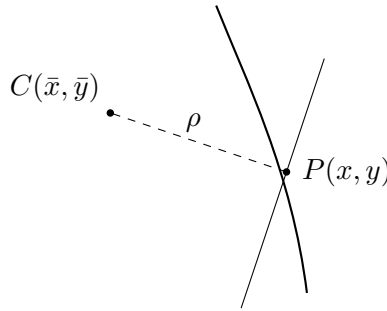
Hence

$$\rho = \frac{N_r}{D_r} = \frac{r^3 \sec^3 2\theta}{-r^2 \sec^2 2\theta} = |-r \sec 2\theta| = r \sec 2\theta \quad (\rho \text{ is positive}).$$

Since $r^2 = a^2 \sec 2\theta$, i.e. $\sec 2\theta = \frac{r^2}{a^2}$,

$$\rho = r \cdot \frac{r^2}{a^2} = \frac{r^3}{a^2}.$$

3 Centre of Curvature

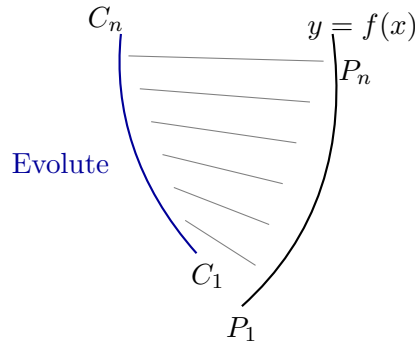


- Consider a point P on the curve $y = f(x)$.
- Draw the tangent at the point P .
- Correspondingly, draw the normal line at P .
- If ρ is the radius of curvature at P , locate the point C on the normal at P (on the concave side) such that $CP = \rho$.
- The point C is called the *centre of curvature* of the curve $y = f(x)$ at the point P .
- Hence, different points of the curve have different centres of curvature.

The centre of curvature $C(\bar{x}, \bar{y})$ of the curve $y = f(x)$ at the point $P(x, y)$ is given by

$$\bar{x} = x - \frac{y_1}{y_2} (1 + y_1^2), \quad \bar{y} = y + \frac{1 + y_1^2}{y_2}, \quad y_1 = \left. \frac{dy}{dx} \right|_{(x,y)}, \quad y_2 = \left. \frac{d^2y}{dx^2} \right|_{(x,y)}.$$

4 Evolute and Involute



- Let $P_1, P_2, \dots, P_n$ be points of the curve $y = f(x)$ and let $C_1, C_2, \dots, C_n$ be their respective centres of curvature.
- These centres of curvature trace another curve, known as the *evolute* of the given curve. That is,

the locus of the centres of curvature of a curve is called the evolute of the curve.

- The given curve $y = f(x)$ is known as the *involute* of the evolute.

Finding the evolute. For the given curve $y = f(x)$, find $\bar{x}$ and $\bar{y}$ (the centre of curvature), and eliminate the parameter to obtain the relationship between $\bar{x}$ and $\bar{y}$. This relation is the evolute.

4.1 Problems on finding the evolute

Problem 4.1. Find the evolute of the parabola $y^2 = 4ax$.

Solution. Differentiating with respect to x : $2y \frac{dy}{dx} = 4a$, so

$$y_1 = \frac{2a}{y}, \quad y_2 = -2a y^{-2} \frac{dy}{dx} = -\frac{2a}{y^2} \cdot \frac{2a}{y} = -\frac{4a^2}{y^3}.$$

Finding $\bar{x}$.

$$\begin{aligned} \bar{x} &= x - \frac{y_1}{y_2} (1 + y_1^2) = x - \frac{2a/y}{-4a^2/y^3} \left(1 + \left(\frac{2a}{y} \right)^2 \right) = x + \frac{2a}{y} \cdot \frac{y^3}{4a^2} \left(1 + \frac{4a^2}{y^2} \right) \\ &= x + \frac{y^2}{2a} \cdot \frac{y^2 + 4a^2}{y^2} = x + \frac{y^2 + 4a^2}{2a} = x + \frac{4ax + 4a^2}{2a} \quad (y^2 = 4ax) \\ &= x + 2x + 2a, \end{aligned}$$

$$\boxed{\bar{x} = 3x + 2a} \quad \Rightarrow \quad x = \frac{\bar{x} - 2a}{3}.$$

Finding $\bar{y}$.

$$\begin{aligned} \bar{y} &= y + \frac{1 + y_1^2}{y_2} = y + \frac{1 + (2a/y)^2}{-4a^2/y^3} = y - \left(\frac{y^2 + 4a^2}{y^2} \right) \frac{y^3}{4a^2} = y - \frac{(y^2 + 4a^2)y}{4a^2} \\ &= y - \frac{y^3}{4a^2} - \frac{4a^2 y}{4a^2} = y - \frac{y^3}{4a^2} - y, \end{aligned}$$

$$\bar{y} = -\frac{y^3}{4a^2}.$$

Eliminating. Squaring, and using $y^6 = (y^2)^3$ and $y^2 = 4ax$,

$$\bar{y}^2 = \frac{y^6}{16a^4} = \frac{(4ax)^3}{16a^4} = \frac{64a^3x^3}{16a^4} = \frac{4x^3}{a} = \frac{4}{a} \left(\frac{\bar{x} - 2a}{3} \right)^3 = \frac{4}{a} \cdot \frac{1}{27} (\bar{x} - 2a)^3.$$

$$27a\bar{y}^2 = 4(\bar{x} - 2a)^3$$

Generally, $27ay^2 = 4(x - 2a)^3$ is the evolute of the given parabola.

Problem 4.2. Find the evolute of the parabola $x^2 = 4ay$.

Solution. $y = \frac{x^2}{4a}$. Differentiating with respect to x :

$$y_1 = \frac{2x}{4a} = \frac{x}{2a}, \quad y_2 = \frac{1}{2a}.$$

Finding $\bar{x}$.

$$\bar{x} = x - \frac{y_1}{y_2}(1 + y_1^2) = x - \frac{x}{2a} \cdot 2a \left(1 + \left(\frac{x}{2a} \right)^2 \right) = x - x \left(1 + \frac{x^2}{4a^2} \right) = x - x - \frac{x^3}{4a^2},$$

$$\bar{x} = -\frac{x^3}{4a^2}.$$

Finding $\bar{y}$.

$$\begin{aligned} \bar{y} &= y + \frac{1 + y_1^2}{y_2} = y + \frac{1 + (x/2a)^2}{1/2a} = y + \left(1 + \frac{x^2}{4a^2} \right) 2a = y + \frac{4a^2 + x^2}{4a^2} \cdot 2a = y + \frac{4a^2 + x^2}{2a} \\ &= y + \frac{4a^2 + 4ay}{2a} \quad (x^2 = 4ay) \\ &= y + 2a + 2y, \end{aligned}$$

$$\bar{y} = 3y + 2a \quad \Rightarrow \quad y = \frac{\bar{y} - 2a}{3}.$$

Eliminating. Squaring $\bar{x}$,

$$\bar{x}^2 = \frac{x^6}{16a^4} = \frac{(4ay)^3}{16a^4} = \frac{64a^3y^3}{16a^4} = \frac{4y^3}{a} = \frac{4}{a} \left(\frac{\bar{y} - 2a}{3} \right)^3 = \frac{4}{a} \cdot \frac{(\bar{y} - 2a)^3}{27}.$$

$$27a\bar{x}^2 = 4(\bar{y} - 2a)^3$$

Generally, $27ax^2 = 4(y - 2a)^3$ is the evolute of the given parabola.

Problem 4.3. Find the evolute of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Solution. Take the parametric form of the ellipse: $x = a \cos \theta$, $y = b \sin \theta$. Then

$$\frac{dx}{d\theta} = -a \sin \theta, \quad \frac{dy}{d\theta} = b \cos \theta, \quad \frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{b \cos \theta}{-a \sin \theta} = -\frac{b}{a} \cot \theta.$$

So $y_1 = -\frac{b}{a} \cot \theta$. Differentiating with respect to x :

$$y_2 = -\frac{b}{a} (-\operatorname{cosec}^2 \theta) \frac{d\theta}{dx} = \frac{b}{a} \operatorname{cosec}^2 \theta \cdot \frac{1}{-a \sin \theta} \Rightarrow y_2 = -\frac{b}{a^2} \operatorname{cosec}^3 \theta.$$

Finding $\bar{x}$.

$$\begin{aligned} \bar{x} &= x - \frac{y_1}{y_2} (1 + y_1^2) = a \cos \theta - \frac{-\frac{b}{a} \cot \theta}{-\frac{b}{a^2} \operatorname{cosec}^3 \theta} \left(1 + \frac{b^2 \cot^2 \theta}{a^2} \right) = a \cos \theta - \frac{a \cot \theta}{\operatorname{cosec}^3 \theta} \left(1 + \frac{b^2 \cot^2 \theta}{a^2} \right) \\ &= a \cos \theta - a \frac{\cos \theta / \sin \theta}{1 / \sin^3 \theta} \left(1 + \frac{b^2 \cos^2 \theta}{a^2 \sin^2 \theta} \right) = a \cos \theta - a \cos \theta \sin^2 \theta \left(\frac{a^2 \sin^2 \theta + b^2 \cos^2 \theta}{a^2 \sin^2 \theta} \right) \\ &= a \cos \theta - \frac{\cos \theta (a^2 \sin^2 \theta + b^2 \cos^2 \theta)}{a} = a \cos \theta - \left(\frac{a^2 \sin^2 \theta \cos \theta}{a} + \frac{b^2 \cos^3 \theta}{a} \right), \\ \bar{x} &= \frac{a^2 \cos \theta - a^2 \sin^2 \theta \cos \theta - b^2 \cos^3 \theta}{a}. \end{aligned}$$

Hence (converting $\sin^2 \theta$ to $\cos \theta$)

$$a\bar{x} = a^2 \cos \theta - a^2 (1 - \cos^2 \theta) \cos \theta - b^2 \cos^3 \theta = a^2 \cos \theta - a^2 \cos \theta + a^2 \cos^3 \theta - b^2 \cos^3 \theta,$$

$$\boxed{a\bar{x} = (a^2 - b^2) \cos^3 \theta}.$$

Finding $\bar{y}$.

$$\begin{aligned} \bar{y} &= y + \frac{1 + y_1^2}{y_2} = y + \frac{1 + \frac{b^2}{a^2} \cot^2 \theta}{-\frac{b}{a^2} \operatorname{cosec}^3 \theta} = y - \frac{(a^2 + b^2 \cot^2 \theta)/a^2}{b \operatorname{cosec}^3 \theta / a^2} = y - \frac{a^2 + b^2 \cot^2 \theta}{b \operatorname{cosec}^3 \theta} \\ &= y - \frac{a^2 + b^2 \frac{\cos^2 \theta}{\sin^2 \theta}}{\frac{b}{\sin^3 \theta}} = y - \frac{(a^2 \sin^2 \theta + b^2 \cos^2 \theta) / \sin^2 \theta}{b / \sin^3 \theta} = y - \frac{(a^2 \sin^2 \theta + b^2 \cos^2 \theta) \sin \theta}{b} \\ &= b \sin \theta - \frac{a^2 \sin^3 \theta + b^2 \cos^2 \theta \sin \theta}{b}. \end{aligned}$$

Hence (converting $\cos^2 \theta$ to $\sin \theta$)

$$b\bar{y} = b^2 \sin \theta - a^2 \sin^3 \theta - b^2 (1 - \sin^2 \theta) \sin \theta = b^2 \sin \theta - a^2 \sin^3 \theta - b^2 \sin \theta + b^2 \sin^3 \theta,$$

$$\boxed{b\bar{y} = (b^2 - a^2) \sin^3 \theta = -(a^2 - b^2) \sin^3 \theta}.$$

Eliminating θ .

$$\cos \theta = \left(\frac{a\bar{x}}{a^2 - b^2} \right)^{1/3} \quad (1), \quad -\sin \theta = \left(\frac{b\bar{y}}{a^2 - b^2} \right)^{1/3} \quad (2).$$

Squaring and adding (1) and (2), using $\sin^2 \theta + \cos^2 \theta = 1$:

$$1 = \left(\frac{b\bar{y}}{a^2 - b^2} \right)^{2/3} + \left(\frac{a\bar{x}}{a^2 - b^2} \right)^{2/3} = \frac{(a\bar{x})^{2/3} + (b\bar{y})^{2/3}}{(a^2 - b^2)^{2/3}},$$

$$(a\bar{x})^{2/3} + (b\bar{y})^{2/3} = (a^2 - b^2)^{2/3}.$$

Generally, the evolute of the ellipse is

$$(ax)^{2/3} + (by)^{2/3} = (a^2 - b^2)^{2/3}.$$

Problem 4.4. Find the evolute of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

Solution. Take the parametric form $x = a \sec \theta$, $y = b \tan \theta$. Then

$$\frac{dx}{d\theta} = a \tan \theta \sec \theta, \quad \frac{dy}{d\theta} = b \sec^2 \theta, \quad \frac{dy}{dx} = \frac{b \sec^2 \theta}{a \tan \theta \sec \theta} = \frac{b}{a} \sec \theta \cot \theta = \frac{b}{a} \cdot \frac{1}{\cos \theta} \cdot \frac{\cos \theta}{\sin \theta}.$$

So $y_1 = \frac{b}{a} \operatorname{cosec} \theta$. Differentiating with respect to x :

$$y_2 = \frac{b}{a} (-\cot \theta \operatorname{cosec} \theta) \frac{d\theta}{dx} = -\frac{b}{a} \cot \theta \operatorname{cosec} \theta \cdot \frac{1}{a \tan \theta \sec \theta} = -\frac{b}{a} \cdot \frac{\cos \theta}{\sin \theta} \cdot \frac{1}{\sin \theta} \cdot \frac{1}{a} \cdot \frac{\cos \theta}{\sin \theta} \cdot \cos \theta,$$

$$y_2 = -\frac{b}{a^2} \cot^3 \theta.$$

Finding $\bar{x}$.

$$\begin{aligned} \bar{x} &= x - \frac{y_1}{y_2} (1 + y_1^2) = a \sec \theta - \frac{\frac{b}{a} \operatorname{cosec} \theta}{-\frac{b}{a^2} \cot^3 \theta} \left(1 + \frac{b^2}{a^2} \operatorname{cosec}^2 \theta \right) \\ &= a \sec \theta + \frac{a \operatorname{cosec} \theta}{\cot^3 \theta} \left(1 + \frac{b^2}{a^2} \operatorname{cosec}^2 \theta \right) \\ &= a \sec \theta + a \cdot \frac{1}{\sin \theta} \cdot \frac{\sin^3 \theta}{\cos^3 \theta} \left(1 + \frac{b^2}{a^2} \cdot \frac{1}{\sin^2 \theta} \right) \\ &= a \sec \theta + a \frac{\sin^2 \theta}{\cos^3 \theta} \left(\frac{a^2 \sin^2 \theta + b^2}{a^2 \sin^2 \theta} \right) \\ &= a \sec \theta + \frac{a \sin^2 \theta}{\cos^3 \theta} + \frac{b^2}{a} \cdot \frac{1}{\cos^3 \theta} \\ &= a \sec \theta + a \tan^2 \theta \sec \theta + \frac{b^2}{a} \sec^3 \theta. \end{aligned}$$

So $a\bar{x} = a^2 \sec \theta + a^2 \tan^2 \theta \sec \theta + b^2 \sec^3 \theta$. Using $\tan^2 \theta = \sec^2 \theta - 1$,

$$a\bar{x} = a^2 \sec \theta + a^2 (\sec^2 \theta - 1) \sec \theta + b^2 \sec^3 \theta = a^2 \sec \theta + a^2 \sec^3 \theta - a^2 \sec \theta + b^2 \sec^3 \theta,$$

$$\boxed{a\bar{x} = (a^2 + b^2) \sec^3 \theta}^1$$

Finding $\bar{y}$.

$$\begin{aligned} \bar{y} &= y + \frac{1 + y_1^2}{y_2} = b \tan \theta + \frac{1 + \frac{b^2}{a^2} \operatorname{cosec}^2 \theta}{-\frac{b}{a^2} \cot^3 \theta} \\ &= b \tan \theta - \frac{1 + \frac{b^2}{a^2} \cdot \frac{1}{\sin^2 \theta}}{\frac{b}{a^2} \cdot \frac{\cos^3 \theta}{\sin^3 \theta}} \end{aligned}$$

¹The handwritten notes record this as $(a^2 - b^2) \sec^3 \theta$; the working above gives $a^2 \sec^3 \theta + b^2 \sec^3 \theta = (a^2 + b^2) \sec^3 \theta$, which is the standard result and is used here.

$$\begin{aligned}
&= b \tan \theta - \frac{(a^2 \sin^2 \theta + b^2)/a^2 \sin^2 \theta}{b \cos^3 \theta / a^2 \sin^3 \theta} = b \tan \theta - \frac{(a^2 \sin^2 \theta + b^2) \sin \theta}{b \cos^3 \theta} \\
&= b \tan \theta - \left(\frac{a^2 \sin^3 \theta}{b \cos^3 \theta} + \frac{b^2 \sin \theta}{b \cos^3 \theta} \right) \\
&= b \tan \theta - \left(\frac{a^2}{b} \tan^3 \theta + b \tan \theta \sec^2 \theta \right) \\
&= b \tan \theta - \left(\frac{a^2}{b} \tan^3 \theta + b \tan \theta (1 + \tan^2 \theta) \right) \\
&= b \tan \theta - \frac{a^2}{b} \tan^3 \theta - b \tan \theta - b \tan^3 \theta,
\end{aligned}$$

$$\bar{y} = -\frac{a^2}{b} \tan^3 \theta - b \tan^3 \theta \Rightarrow b\bar{y} = -a^2 \tan^3 \theta - b^2 \tan^3 \theta, \quad \boxed{b\bar{y} = -(a^2 + b^2) \tan^3 \theta}.$$

Eliminating θ .

$$\left(\frac{a\bar{x}}{a^2 + b^2} \right)^{1/3} = \sec \theta, \quad \left(\frac{b\bar{y}}{a^2 + b^2} \right)^{1/3} = -\tan \theta.$$

Squaring and subtracting, using $\sec^2 \theta - \tan^2 \theta = 1$:

$$1 = \left(\frac{a\bar{x}}{a^2 + b^2} \right)^{2/3} - \left(\frac{b\bar{y}}{a^2 + b^2} \right)^{2/3} \Rightarrow (a\bar{x})^{2/3} - (b\bar{y})^{2/3} = (a^2 + b^2)^{2/3}.$$

Generally, the evolute of the hyperbola is

$$\boxed{(ax)^{2/3} - (by)^{2/3} = (a^2 + b^2)^{2/3}}.$$

Problem 4.5. Find the evolute of the astroid $x^{2/3} + y^{2/3} = a^{2/3}$.

Solution. Parametric equations: $x = a \cos^3 \theta$, $y = a \sin^3 \theta$.

$$\frac{dx}{d\theta} = a \cdot 3 \cos^2 \theta (-\sin \theta), \quad \frac{dy}{d\theta} = a \cdot 3 \sin^2 \theta \cos \theta,$$

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{3a \sin^2 \theta \cos \theta}{-3a \sin \theta \cos^2 \theta} = -\frac{\sin \theta}{\cos \theta} = -\tan \theta.$$

So $y_1 = -\tan \theta$. Differentiating with respect to x :

$$y_2 = -\sec^2 \theta \frac{d\theta}{dx} = -\sec^2 \theta \cdot \frac{1}{-3a \sin \theta \cos^2 \theta} = \frac{1}{\cos^2 \theta} \cdot \frac{1}{3a \sin \theta \cos^2 \theta} = \frac{1}{3a \sin \theta \cos^4 \theta},$$

$$y_2 = \frac{1}{3a} \sec^4 \theta \operatorname{cosec} \theta.$$

Finding $\bar{x}$. Using $1 + \tan^2 \theta = \sec^2 \theta = \frac{1}{\cos^2 \theta}$,

$$\begin{aligned}
\bar{x} &= x - \frac{y_1}{y_2} (1 + y_1^2) = a \cos^3 \theta - \frac{-\tan \theta}{\frac{1}{3a} \sec^4 \theta \operatorname{cosec} \theta} (1 + (-\tan \theta)^2) \\
&= a \cos^3 \theta + 3a \frac{\sin \theta / \cos \theta}{\frac{1}{\cos^4 \theta} \cdot \frac{1}{\sin \theta}} (1 + \tan^2 \theta) = a \cos^3 \theta + 3a \cdot \frac{\sin \theta}{\cos \theta} \cdot \cos^4 \theta \cdot \sin \theta \cdot \frac{1}{\cos^2 \theta},
\end{aligned}$$

$$\boxed{\bar{x} = a \cos^3 \theta + 3a \cos \theta \sin^2 \theta} \Rightarrow \frac{\bar{x}}{a} = \cos^3 \theta + 3 \cos \theta \sin^2 \theta.$$

Finding $\bar{y}$.

$$\begin{aligned}\bar{y} &= y + \frac{1 + y_1^2}{y_2} = a \sin^3 \theta + \frac{1 + (-\tan \theta)^2}{\frac{1}{3a} \sec^4 \theta \operatorname{cosec} \theta} = a \sin^3 \theta + 3a \frac{1 + \tan^2 \theta}{\sec^4 \theta \operatorname{cosec} \theta} \\ &= a \sin^3 \theta + 3a \frac{\sec^2 \theta}{\sec^4 \theta \operatorname{cosec} \theta} = a \sin^3 \theta + 3a \cos^2 \theta \sin \theta, \\ \boxed{\bar{y} = a \sin^3 \theta + 3a \cos^2 \theta \sin \theta} &\Rightarrow \frac{\bar{y}}{a} = \sin^3 \theta + 3 \cos^2 \theta \sin \theta.\end{aligned}$$

Eliminating θ . Adding and subtracting,

$$\frac{\bar{x}}{a} + \frac{\bar{y}}{a} = (\cos \theta + \sin \theta)^3, \quad \frac{\bar{x}}{a} - \frac{\bar{y}}{a} = (\cos \theta - \sin \theta)^3,$$

since $\cos^3 \theta + \sin^3 \theta + 3 \cos^2 \theta \sin \theta + 3 \cos \theta \sin^2 \theta = (\cos \theta + \sin \theta)^3$, and similarly for the difference. Therefore

$$\left(\frac{\bar{x}}{a} + \frac{\bar{y}}{a}\right)^{1/3} = \cos \theta + \sin \theta, \quad \left(\frac{\bar{x}}{a} - \frac{\bar{y}}{a}\right)^{1/3} = \cos \theta - \sin \theta.$$

Squaring and adding,

$$\left(\frac{\bar{x}}{a} + \frac{\bar{y}}{a}\right)^{2/3} + \left(\frac{\bar{x}}{a} - \frac{\bar{y}}{a}\right)^{2/3} = \cos^2 \theta + \sin^2 \theta + 2 \sin \theta \cos \theta + \cos^2 \theta + \sin^2 \theta - 2 \sin \theta \cos \theta = 2.$$

General form of the evolute:

$$\boxed{(x + y)^{2/3} + (x - y)^{2/3} = 2a^{2/3}}.$$

Problem 4.6. Show that the evolute of the curve $x = a(\cos t + t \sin t)$, $y = a(\sin t - t \cos t)$ is a circle.

Solution.

$$\begin{aligned}\frac{dx}{dt} &= a(-\sin t + t \cos t + \sin t \cdot 1) = at \cos t, \\ \frac{dy}{dt} &= a(\cos t - (t(-\sin t) + \cos t)) = a(\cos t + t \sin t - \cos t) = at \sin t, \\ \frac{dy}{dx} &= \frac{dy/dt}{dx/dt} = \frac{at \sin t}{at \cos t} = \tan t \Rightarrow y_1 = \tan t, \\ y_2 &= \sec^2 t \frac{dt}{dx} = \sec^2 t \cdot \frac{1}{at \cos t} \Rightarrow y_2 = \frac{1}{at} \sec^3 t.\end{aligned}$$

Finding $\bar{x}$.

$$\begin{aligned}\bar{x} &= x - \frac{y_1}{y_2}(1 + y_1^2) = a(\cos t + t \sin t) - \frac{\tan t}{\sec^3 t/at}(1 + \tan^2 t) = a(\cos t + t \sin t) - at \frac{\tan t}{\sec^3 t} \cdot \sec^2 t \\ &= a \cos t + at \sin t - at \frac{\sin t}{\cos t} \cdot \cos t \Rightarrow \boxed{\bar{x} = a \cos t}.\end{aligned}$$

Finding $\bar{y}$.

$$\begin{aligned}\bar{y} &= y + \frac{1 + y_1^2}{y_2} = a(\sin t - t \cos t) + \frac{1 + \tan^2 t}{\sec^3 t/at} \\ &= a \sin t - at \cos t + at \frac{\sec^2 t}{\sec^3 t} = a \sin t - at \cos t + at \cos t \Rightarrow \boxed{\bar{y} = a \sin t}.\end{aligned}$$

Squaring and adding: $\bar{x}^2 + \bar{y}^2 = a^2(\sin^2 t + \cos^2 t) = a^2$. Hence the evolute is

$$\boxed{x^2 + y^2 = a^2},$$

which is the equation of a circle.

Problem 4.7. Show that the evolute of the cycloid $x = a(\theta - \sin \theta)$, $y = a(1 - \cos \theta)$ is another cycloid $x = a(\theta + \sin \theta)$, $y = -a(1 - \cos \theta)$.

Solution.

$$\frac{dx}{d\theta} = a(1 - \cos \theta), \quad \frac{dy}{d\theta} = a(0 - (-\sin \theta)) = a \sin \theta, \quad \frac{dy}{dx} = \frac{a \sin \theta}{a(1 - \cos \theta)} = \frac{\sin \theta}{1 - \cos \theta}.$$

$$y_1 = \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \sin^2 \frac{\theta}{2}} = \cot \frac{\theta}{2}.$$

$$y_2 = -\operatorname{cosec}^2 \frac{\theta}{2} \cdot \frac{1}{2} \cdot \frac{d\theta}{dx} = -\operatorname{cosec}^2 \frac{\theta}{2} \cdot \frac{1}{2} \cdot \frac{1}{a(1 - \cos \theta)} = -\frac{1}{a} \operatorname{cosec}^2 \frac{\theta}{2} \cdot \frac{1}{2} \cdot \frac{1}{2 \sin^2 \frac{\theta}{2}},$$

$$y_2 = -\frac{1}{4a} \operatorname{cosec}^4 \frac{\theta}{2}.$$

Finding $\bar{x}$. Using $1 + \cot^2 \frac{\theta}{2} = \operatorname{cosec}^2 \frac{\theta}{2}$,

$$\begin{aligned} \bar{x} &= x - \frac{y_1}{y_2} (1 + y_1^2) = a(\theta - \sin \theta) - \frac{\cot \frac{\theta}{2}}{-\frac{1}{4a} \operatorname{cosec}^4 \frac{\theta}{2}} \left(1 + \cot^2 \frac{\theta}{2} \right) \\ &= a(\theta - \sin \theta) + 4a \frac{\cot \frac{\theta}{2}}{\operatorname{cosec}^4 \frac{\theta}{2}} \operatorname{cosec}^2 \frac{\theta}{2} \\ &= a(\theta - \sin \theta) + 4a \frac{\cot \frac{\theta}{2}}{\operatorname{cosec}^2 \frac{\theta}{2}} = a(\theta - \sin \theta) + 4a \cdot \frac{\cos \frac{\theta}{2}}{\sin \frac{\theta}{2}} \cdot \sin^2 \frac{\theta}{2} \\ &= a(\theta - \sin \theta) + 2a \left(2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} \right) \\ &= a(\theta - \sin \theta) + 2a \sin \theta = a\theta - a \sin \theta + 2a \sin \theta = a\theta + a \sin \theta, \end{aligned}$$

$$\boxed{\bar{x} = a(\theta + \sin \theta)}.$$

Finding $\bar{y}$. Using $2 \sin^2 \frac{\theta}{2} = 1 - \cos \theta$,

$$\begin{aligned} \bar{y} &= y + \frac{1 + y_1^2}{y_2} = a(1 - \cos \theta) + \frac{1 + \cot^2 \frac{\theta}{2}}{-\frac{1}{4a} \operatorname{cosec}^4 \frac{\theta}{2}} = a(1 - \cos \theta) - 4a \frac{\operatorname{cosec}^2 \frac{\theta}{2}}{\operatorname{cosec}^4 \frac{\theta}{2}} \\ &= a(1 - \cos \theta) - 4a \sin^2 \frac{\theta}{2} = a(1 - \cos \theta) - 2a \left(2 \sin^2 \frac{\theta}{2} \right) = a(1 - \cos \theta) - 2a(1 - \cos \theta) \\ &= a - a \cos \theta - 2a + 2a \cos \theta = -a + a \cos \theta, \end{aligned}$$

$$\boxed{\bar{y} = -a(1 - \cos \theta)}.$$

Generally, the evolute is $x = a(\theta + \sin \theta)$, $y = -a(1 - \cos \theta)$. Hence the evolute of the given cycloid is another cycloid.

Problem 4.8. Show that the evolute of the tractrix $x = a \left(\cos \theta + \ln \tan \frac{\theta}{2} \right)$, $y = a \sin \theta$ is the catenary $y = a \cosh \left(\frac{x}{a} \right)$.

Solution.

$$\begin{aligned} \frac{dx}{d\theta} &= a \left(-\sin \theta + \frac{1}{\tan \frac{\theta}{2}} \cdot \sec^2 \frac{\theta}{2} \cdot \frac{1}{2} \right) = -a \sin \theta + \frac{a}{2} \cdot \frac{\cos \frac{\theta}{2}}{\sin \frac{\theta}{2}} \cdot \frac{1}{\cos^2 \frac{\theta}{2}} = -a \sin \theta + \frac{a}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} \\ &= -a \sin \theta + \frac{a}{\sin \theta} = \frac{a - a \sin^2 \theta}{\sin \theta}, \quad \frac{dy}{d\theta} = a \cos \theta. \\ \frac{dy}{dx} &= \frac{dy/d\theta}{dx/d\theta} = \frac{a \cos \theta \sin \theta}{a - a \sin^2 \theta} = \frac{a \sin \theta \cos \theta}{a(1 - \sin^2 \theta)} = \frac{\sin \theta \cos \theta}{\cos^2 \theta} = \tan \theta. \end{aligned}$$

So $y_1 = \tan \theta$, and

$$y_2 = \sec^2 \theta \frac{d\theta}{dx} = \sec^2 \theta \cdot \frac{\sin \theta}{a(1 - \sin^2 \theta)} = \sec^2 \theta \cdot \frac{\sin \theta}{a \cos^2 \theta} \Rightarrow y_2 = \frac{1}{a} \sec^4 \theta \sin \theta.$$

Finding $\bar{x}$.

$$\begin{aligned} \bar{x} &= x - \frac{y_1}{y_2} (1 + y_1^2) = a \cos \theta + a \ln \tan \frac{\theta}{2} - \frac{\tan \theta}{\sec^4 \theta \sin \theta / a} (1 + \tan^2 \theta) \\ &= a \cos \theta + a \ln \tan \frac{\theta}{2} - a \frac{\tan \theta \sec^2 \theta}{\sin \theta \sec^4 \theta} = a \cos \theta + a \ln \tan \frac{\theta}{2} - a \cdot \frac{\sin \theta}{\cos \theta} \cdot \frac{1}{\sin \theta} \cdot \cos^2 \theta \\ &= a \cos \theta + a \ln \tan \frac{\theta}{2} - a \cos \theta, \end{aligned}$$

$$\boxed{\bar{x} = a \ln \tan \frac{\theta}{2}}.$$

Finding $\bar{y}$.

$$\begin{aligned} \bar{y} &= y + \frac{1 + y_1^2}{y_2} = a \sin \theta + \frac{1 + \tan^2 \theta}{\sec^4 \theta \sin \theta / a} = a \sin \theta + a \frac{\sec^2 \theta}{\sec^4 \theta \sin \theta} \\ &= a \sin \theta + \frac{a \cos^2 \theta}{\sin \theta} = \frac{a \sin^2 \theta + a \cos^2 \theta}{\sin \theta}, \end{aligned}$$

$$\boxed{\bar{y} = \frac{a}{\sin \theta}}.$$

Eliminating θ . From $\bar{x} = a \ln \tan \frac{\theta}{2}$ we get $\tan \frac{\theta}{2} = e^{\bar{x}/a}$. Then, using $\sin \theta = \frac{2 \tan \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}}$,

$$\begin{aligned} \bar{y} &= \frac{a}{\sin \theta} = \frac{a(1 + \tan^2 \frac{\theta}{2})}{2 \tan \frac{\theta}{2}} = \frac{a(1 + e^{2\bar{x}/a})}{2e^{\bar{x}/a}} = \frac{a}{2e^{\bar{x}/a}} + \frac{ae^{2\bar{x}/a}}{2e^{\bar{x}/a}} \\ &= \frac{a}{2} \left(\frac{1}{e^{\bar{x}/a}} + e^{\bar{x}/a} \right) = \frac{a}{2} \left(e^{-\bar{x}/a} + e^{\bar{x}/a} \right) = a \left(\frac{e^{\bar{x}/a} + e^{-\bar{x}/a}}{2} \right). \end{aligned}$$

Hence the evolute is

$$\boxed{y = a \cosh \left(\frac{x}{a} \right)},$$

which is a catenary.

Problem 4.9. If ρ_1 and ρ_2 are the radii of curvature at corresponding points of the cycloid and its evolute, prove that $\rho_1^2 + \rho_2^2$ is a constant.

Solution. From the previous problem:

Cycloid (1): $x = a(\theta - \sin \theta)$, $y = a(1 - \cos \theta)$. Evolute of the cycloid (2): $x = a(\theta + \sin \theta)$, $y = -a(1 - \cos \theta)$.

For curve (1) (from the previous problem), $y_1 = \cot \frac{\theta}{2}$ and $y_2 = -\frac{1}{4a} \operatorname{cosec}^4 \frac{\theta}{2}$. So

$$\rho_1 = \frac{(1 + y_1^2)^{3/2}}{y_2} = \frac{(1 + \cot^2 \frac{\theta}{2})^{3/2}}{-\frac{1}{4a} \operatorname{cosec}^4 \frac{\theta}{2}} = -4a \frac{(\operatorname{cosec}^2 \frac{\theta}{2})^{3/2}}{\operatorname{cosec}^4 \frac{\theta}{2}} = -4a \frac{\operatorname{cosec}^3 \frac{\theta}{2}}{\operatorname{cosec}^4 \frac{\theta}{2}} = \frac{-4a}{\operatorname{cosec} \frac{\theta}{2}},$$

$$\rho_1 = -4a \sin \frac{\theta}{2} \quad (\text{its magnitude } 4a \sin \frac{\theta}{2} \text{ is taken}).$$

For curve (2). Using $1 + \cos \theta = 2 \cos^2 \frac{\theta}{2}$,

$$\frac{dx}{d\theta} = a(1 + \cos \theta), \quad \frac{dy}{d\theta} = -a(0 - (-\sin \theta)) = -a \sin \theta,$$

$$\frac{dy}{dx} = \frac{-a \sin \theta}{a(1 + \cos \theta)} = -\frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \cos^2 \frac{\theta}{2}} = -\tan \frac{\theta}{2} \Rightarrow y_1 = -\tan \frac{\theta}{2}.$$

$$y_2 = -\sec^2 \frac{\theta}{2} \cdot \frac{1}{2} \cdot \frac{d\theta}{dx} = -\sec^2 \frac{\theta}{2} \cdot \frac{1}{2} \cdot \frac{1}{a(1 + \cos \theta)} = -\frac{1}{4a} \sec^2 \frac{\theta}{2} \cdot \sec^2 \frac{\theta}{2} \Rightarrow y_2 = -\frac{1}{4a} \sec^4 \frac{\theta}{2}.$$

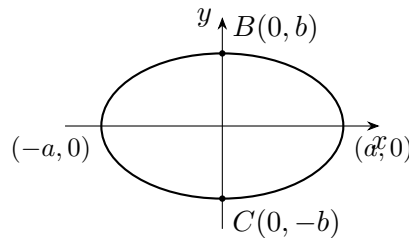
$$\rho_2 = \frac{(1 + \tan^2 \frac{\theta}{2})^{3/2}}{-\frac{1}{4a} \sec^4 \frac{\theta}{2}} = -4a \frac{(\sec^2 \frac{\theta}{2})^{2 \cdot \frac{3}{2}}}{\sec^4 \frac{\theta}{2}} = \frac{-4a}{\sec \frac{\theta}{2}} \Rightarrow \rho_2 = -4a \cos \frac{\theta}{2}.$$

Therefore

$$\rho_1^2 + \rho_2^2 = 16a^2 \sin^2 \frac{\theta}{2} + 16a^2 \cos^2 \frac{\theta}{2} = 16a^2 = \text{constant}.$$

Problem 4.10. If the centre of curvature of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at one end of the minor axis lies at the other end, prove that the eccentricity of the ellipse is $\frac{1}{\sqrt{2}}$.

Solution.



The centre of curvature at $B(0, b)$ lies at $C(0, -b)$. Since ρ is the distance from the point to its centre of curvature,

$$\rho = BC = 2b \quad \text{at the point } (0, b). \quad (1)$$

Now we find ρ for the ellipse. Differentiating with respect to x :

$$\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{2x}{a^2} \cdot \frac{b^2}{2y} \Rightarrow y_1 = -\frac{b^2}{a^2} \cdot \frac{x}{y}, \quad y_1|_{(0,b)} = 0.$$

Differentiating again with respect to x ,

$$y_2 = -\frac{b^2}{a^2} \left(\frac{y(1) - x \frac{dy}{dx}}{y^2} \right).$$

At $(0, b)$:

$$y_2 = -\frac{b^2}{a^2} \left(\frac{b-0}{b^2} \right) = -\frac{b^2}{a^2} \cdot \frac{b}{b^2} \Rightarrow y_2 = -\frac{b}{a^2}.$$

Hence

$$\rho = \frac{(1 + y_1^2)^{3/2}}{y_2} = \frac{(1 + 0)^{3/2}}{-b/a^2} = \left| -\frac{a^2}{b} \right| \Rightarrow \rho = \frac{a^2}{b} \quad (\rho \text{ is positive}).$$

From (1), $2b = \frac{a^2}{b}$, i.e. $a^2 = 2b^2$. The eccentricity of the ellipse is given by

$$e^2 = \frac{a^2 - b^2}{a^2} = \frac{2b^2 - b^2}{2b^2} = \frac{b^2}{2b^2} = \frac{1}{2} \Rightarrow e = \frac{1}{\sqrt{2}} \quad (e > 0).$$

5 Circle of Curvature

Definition 5.1. A circle whose centre is the centre of curvature and whose radius is the radius of curvature of a given curve (at a point) is known as the *circle of curvature* of the given curve at that point.

Hence, for a given curve $y = f(x)$, the equation of its circle of curvature is

$$(x - \bar{x})^2 + (y - \bar{y})^2 = \rho^2,$$

where $(\bar{x}, \bar{y})$ is the centre of curvature and ρ is the radius of curvature of $y = f(x)$.

5.1 Problems

Problem 5.1. Find the circle of curvature of the parabola $y^2 = 12x$ at the point $(3, 6)$.

Solution. $2y \frac{dy}{dx} = 12 \Rightarrow y_1 = \frac{6}{y}$; at $(3, 6)$, $y_1 = \frac{6}{6} = 1$.

$$y_2 = 6(-y^{-2}) \frac{dy}{dx} = -\frac{6}{y^2} \cdot \frac{6}{y} = -\frac{36}{y^3}; \text{ at } (3, 6), y_2 = -\frac{36}{36 \cdot 6} = -\frac{1}{6}.$$

$$\rho = \frac{(1 + y_1^2)^{3/2}}{y_2} = \frac{(1 + 1)^{3/2}}{-1/6} = -6 \cdot 2^{3/2} = -6 \cdot 2\sqrt{2} = -12\sqrt{2}, \quad \text{so } \rho = 12\sqrt{2}, \quad \rho^2 = 288.$$

$$\bar{x} = x - \frac{y_1}{y_2}(1 + y_1^2) = 3 - \frac{1}{-1/6}(1 + 1^2) = 3 + 6(2) = 15,$$

$$\bar{y} = y + \frac{1 + y_1^2}{y_2} = 6 + \frac{1 + 1^2}{-1/6} = 6 - 6(2) = -6.$$

The circle of curvature $(x - \bar{x})^2 + (y - \bar{y})^2 = \rho^2$ is

$$(x - 15)^2 + (y + 6)^2 = 288.$$

Problem 5.2. Find the circle of curvature of $\sqrt{x} + \sqrt{y} = \sqrt{a}$ at $\left(\frac{a}{4}, \frac{a}{4}\right)$.

Solution. $x^{1/2} + y^{1/2} = a^{1/2}$. Differentiating,

$$\frac{1}{2}x^{-1/2} + \frac{1}{2}y^{-1/2}\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{\frac{1}{2}x^{-1/2}}{\frac{1}{2}y^{-1/2}} = -\frac{1/\sqrt{x}}{1/\sqrt{y}} \Rightarrow y_1 = -\left(\frac{y}{x}\right)^{1/2} = -\frac{y^{1/2}}{x^{1/2}}.$$

At $\left(\frac{a}{4}, \frac{a}{4}\right)$, $y_1 = -1$. Differentiating y_1 by the quotient rule,

$$y_2 = -\frac{x^{1/2} \cdot \frac{1}{2}y^{-1/2}\frac{dy}{dx} - y^{1/2} \cdot \frac{1}{2}x^{-1/2}}{x}.$$

At $\left(\frac{a}{4}, \frac{a}{4}\right)$ with $\frac{dy}{dx} = -1$:

$$y_2 = -\frac{\frac{1}{2}(-1) - \frac{1}{2}}{a/4} = -\frac{-1}{a/4} \Rightarrow y_2 = \frac{4}{a}.$$

$$\rho = \frac{(1 + y_1^2)^{3/2}}{y_2} = \frac{(1 + 1)^{3/2}}{4/a} = \frac{a}{4} \cdot 2^{3/2} = \frac{a}{4} \cdot 2\sqrt{2} = \frac{a}{\sqrt{2}}.$$

$$\bar{x} = x - \frac{y_1}{y_2}(1 + y_1^2) = \frac{a}{4} - \frac{-1}{4/a}(1 + 1) = \frac{a}{4} + \frac{a}{4} \cdot 2 = \frac{3a}{4}, \quad \bar{y} = y + \frac{1 + y_1^2}{y_2} = \frac{a}{4} + \frac{1 + 1}{4/a} = \frac{a}{4} + \frac{a}{2} = \frac{3a}{4}.$$

The circle of curvature is

$$\left(x - \frac{3a}{4}\right)^2 + \left(y - \frac{3a}{4}\right)^2 = \frac{a^2}{2} \Rightarrow \frac{(4x - 3a)^2}{4^2} + \frac{(4y - 3a)^2}{4^2} = \frac{a^2}{2}$$

$$\Rightarrow (4x - 3a)^2 + (4y - 3a)^2 = \frac{a^2}{2} \cdot 16,$$

$$\boxed{(4x - 3a)^2 + (4y - 3a)^2 = 8a^2}.$$

6 Envelopes

Definition 6.1. The locus of the ultimate points of intersection of the members of a given family of curves is called the *envelope* of the given family of curves.

Method.

- To find the envelope of a given family of curves $f(x, y, \alpha) = 0$, where α is the parameter, differentiate the relation partially with respect to α : $\frac{\partial f}{\partial \alpha} = 0$.
- Now eliminate α between $f(x, y, \alpha) = 0$ and $\frac{\partial f}{\partial \alpha} = 0$; we get the required envelope.

6.1 Problems

Problem 6.1. Find the envelope of the family of straight lines $y = mx + \frac{a}{m}$, where m is the parameter.

Solution. $f = y - \left(mx + \frac{a}{m}\right) = y - mx - \frac{a}{m} = 0$. Differentiating partially with respect to m :

$$\frac{\partial f}{\partial m} = 0 - x - a(-m^{-2}) = 0 \Rightarrow x = \frac{a}{m^2} \Rightarrow m^2 = \frac{a}{x}. \quad (1)$$

Now $y = mx + \frac{a}{m} = \frac{1}{m}(m^2x + a)$. Squaring,

$$y^2 = \frac{1}{m^2} ((m^2x)^2 + a^2 + 2a(m^2x)).$$

Using (1), $m^2 = a/x$:

$$y^2 = \frac{x}{a} (a^2 + a^2 + 2a \cdot a) = \frac{x}{a} \cdot 4a^2, \quad \boxed{y^2 = 4ax} \Rightarrow \text{a parabola.}$$

So the envelope of this family of straight lines is a parabola.

Note. Suppose the given family of curves can be written as a quadratic equation in the parameter α :

$$A\alpha^2 + B\alpha + C = 0. \quad (1)$$

Then $f = A\alpha^2 + B\alpha + C = 0$, and $\frac{\partial f}{\partial \alpha} = 2A\alpha + B = 0 \Rightarrow \alpha = -\frac{B}{2A}$. Substituting in (1),

$$\frac{AB^2}{4A^2} + B\left(-\frac{B}{2A}\right) + C = 0 \Rightarrow \frac{B^2}{4A} - \frac{B^2}{2A} + C = 0 \Rightarrow -\frac{B^2}{4A} + C = 0 \Rightarrow -B^2 + 4AC = 0.$$

For the family of curves $A\alpha^2 + B\alpha + C = 0$, where α is the parameter, the envelope is given by

$$B^2 - 4AC = 0.$$

Problem 6.1 by this method. $y = mx + \frac{a}{m} \Rightarrow my = m^2x + a \Rightarrow m^2x - my + a = 0$, which is of the form $A\alpha^2 + B\alpha + C = 0$ with $A = x$, $B = -y$, $C = a$. Envelope:

$$B^2 - 4AC = 0 \Rightarrow (-y)^2 - 4(x)(a) = 0 \Rightarrow y^2 = 4ax.$$

Problem 6.2. Find the envelope of the family of straight lines $y = mx + \sqrt{1 + m^2}$, where m is the parameter.

Solution. $y - mx = \sqrt{1 + m^2}$. Squaring,

$$y^2 + m^2x^2 - 2mxy = 1 + m^2 \Rightarrow m^2(x^2 - 1) - 2xym + (y^2 - 1) = 0,$$

which is of the form $A\alpha^2 + B\alpha + C = 0$. Envelope: $B^2 - 4AC = 0$,

$$(-2xy)^2 - 4(x^2 - 1)(y^2 - 1) = 0 \Rightarrow 4x^2y^2 - 4(x^2y^2 - x^2 - y^2 + 1) = 0$$

$$\Rightarrow 4x^2y^2 - 4x^2y^2 + 4x^2 + 4y^2 - 4 = 0,$$

$$4x^2 + 4y^2 = 4 \Rightarrow \boxed{x^2 + y^2 = 1},$$

a circle of radius 1 (the unit circle).

Problem 6.3. Find the envelope of the family of curves $\frac{x \cos \alpha}{a} + \frac{y \sin \alpha}{b} = 1$, where α is the parameter.

Solution.

$$\frac{x}{a} \cos \alpha + \frac{y}{b} \sin \alpha = 1. \quad (1)$$

Differentiating partially with respect to α :

$$\frac{x}{a}(-\sin \alpha) + \frac{y}{b}(\cos \alpha) = 0 \Rightarrow -\frac{x}{a} \sin \alpha + \frac{y}{b} \cos \alpha = 0. \quad (2)$$

Eliminating the first term: multiply (1) by $\sin \alpha$ and (2) by $\cos \alpha$ and add:

$$\begin{array}{rcl} \frac{x \sin \alpha \cos \alpha}{a} + \frac{y \sin^2 \alpha}{b} & = & \sin \alpha \\ -\frac{x \sin \alpha \cos \alpha}{a} + \frac{y \cos^2 \alpha}{b} & = & 0 \\ \hline \frac{y}{b}(\sin^2 \alpha + \cos^2 \alpha) & = & \sin \alpha \end{array} \Rightarrow \sin \alpha = \frac{y}{b}. \quad (3)$$

Eliminating the second term: multiply (1) by $\cos \alpha$ and (2) by $\sin \alpha$ and subtract:

$$\begin{array}{rcl} \frac{x \cos^2 \alpha}{a} + \frac{y \sin \alpha \cos \alpha}{b} & = & \cos \alpha \\ -\frac{x \sin^2 \alpha}{a} + \frac{y \sin \alpha \cos \alpha}{b} & = & 0 \\ \hline \frac{x}{a}(\sin^2 \alpha + \cos^2 \alpha) & = & \cos \alpha \end{array} \Rightarrow \cos \alpha = \frac{x}{a}. \quad (4)$$

Substituting (3) and (4) in (1):

$$\frac{x}{a} \left(\frac{x}{a} \right) + \frac{y}{b} \left(\frac{y}{b} \right) = 1 \Rightarrow \boxed{\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1} \quad (\text{an ellipse}).$$

Alternatively, squaring and adding (3) and (4): $\sin^2 \alpha + \cos^2 \alpha = 1 = \frac{x^2}{a^2} + \frac{y^2}{b^2}$.

Problem 6.4. Find the envelope of the family of curves $x + y \sin \theta = a \cos \theta$, where θ is the parameter.

Solution.

$$x + y \sin \theta - a \cos \theta = 0. \quad (1)$$

Differentiating partially with respect to θ : $0 + y \cos \theta = a(-\sin \theta)$, i.e.

$$y \cos \theta + a \sin \theta = 0. \quad (2)$$

Multiply (1) by $\sin \theta$ and (2) by $\cos \theta$, and add:

$$\begin{array}{rcl} x \sin \theta + y \sin^2 \theta - a \sin \theta \cos \theta & = & 0 \\ y \cos^2 \theta + a \sin \theta \cos \theta & = & 0 \\ \hline x \sin \theta + y(\sin^2 \theta + \cos^2 \theta) & = & 0 \end{array} \Rightarrow \sin \theta = -\frac{y}{x}.$$

Multiply (1) by $\cos \theta$ and (2) by $\sin \theta$, and subtract:

$$\begin{array}{rcl} x \cos \theta + y \sin \theta \cos \theta - a \cos^2 \theta & = & 0 \\ y \sin \theta \cos \theta + a \sin^2 \theta & = & 0 \\ \hline x \cos \theta - a(\sin^2 \theta + \cos^2 \theta) & = & 0 \end{array} \Rightarrow \cos \theta = \frac{a}{x}.$$

Substituting in (1):

$$x + y \left(-\frac{y}{x} \right) = a \left(\frac{a}{x} \right) \Rightarrow \frac{x^2 - y^2}{x} = \frac{a^2}{x} \Rightarrow \boxed{x^2 - y^2 = a^2}.$$

Problem 6.5 (Homework). Find the envelope of the family of curves

- (i) $y = mx + \sqrt{a^2m^2 + b^2}$, m being the parameter;
(ii) $x \cos^3 \alpha + y \sin^3 \alpha = a$, α being the parameter.

Solution. (i) $y - mx = \sqrt{a^2m^2 + b^2}$. Squaring,

$$y^2 + m^2x^2 - 2mxy = a^2m^2 + b^2 \Rightarrow m^2(x^2 - a^2) - (2xy)m + (y^2 - b^2) = 0,$$

of the form $A\alpha^2 + B\alpha + C = 0$. Envelope: $B^2 - 4AC = 0$,

$$4x^2y^2 - 4(x^2 - a^2)(y^2 - b^2) = 0 \Rightarrow 4x^2y^2 - 4x^2y^2 + 4x^2b^2 + 4y^2a^2 - 4a^2b^2 = 0,$$

$$x^2b^2 + y^2a^2 = a^2b^2 \Rightarrow \boxed{\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1} \quad (\text{an ellipse}).$$

$$(ii) \quad x \cos^3 \alpha + y \sin^3 \alpha = a \quad (1)$$

Differentiating partially with respect to α :

$$x(3 \cos^2 \alpha)(-\sin \alpha) + y(3 \sin^2 \alpha)(\cos \alpha) = 0 \Rightarrow -3x \cos^2 \alpha \sin \alpha + 3y \sin^2 \alpha \cos \alpha = 0. \quad (2)$$

Multiply (1) by $3 \sin \alpha$ and (2) by $\cos \alpha$, and add:

$$\begin{array}{rcl} 3x \cos^3 \alpha \sin \alpha + 3y \sin^4 \alpha & = & 3a \sin \alpha \\ -3x \cos^3 \alpha \sin \alpha + 3y \sin^2 \alpha \cos^2 \alpha & = & 0 \\ \hline 3y \sin^2 \alpha (\sin^2 \alpha + \cos^2 \alpha) & = & 3a \sin \alpha \end{array} \Rightarrow \sin \alpha = \frac{a}{y}.$$

Multiply (1) by $3 \cos \alpha$ and (2) by $\sin \alpha$, and subtract:

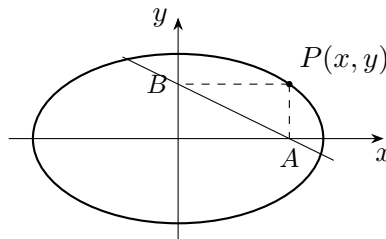
$$\begin{array}{rcl} 3x \cos^4 \alpha + 3y \sin^3 \alpha \cos \alpha & = & 3a \cos \alpha \\ -3x \cos^2 \alpha \sin^2 \alpha + 3y \sin^3 \alpha \cos \alpha & = & 0 \\ \hline 3x \cos^2 \alpha (\cos^2 \alpha + \sin^2 \alpha) & = & 3a \cos \alpha \end{array} \Rightarrow \cos \alpha = \frac{a}{x}.$$

Substituting in (1):

$$x \left(\frac{a^3}{x^3} \right) + y \left(\frac{a^3}{y^3} \right) = a \Rightarrow \frac{a^3}{x^2} + \frac{a^3}{y^2} = a \Rightarrow \frac{a^2}{x^2} + \frac{a^2}{y^2} = 1 \Rightarrow \boxed{\frac{1}{x^2} + \frac{1}{y^2} = \frac{1}{a^2}}.$$

Problem 6.6. From any point on an ellipse, perpendiculars are drawn to the axes and the feet of these perpendiculars are joined. Find the envelope of the family of lines thus formed.

Solution.



Parametric form of the ellipse: $x = a \cos \theta$, $y = b \sin \theta$. The point is $P(a \cos \theta, b \sin \theta)$ and the feet of the perpendiculars are $A = (a \cos \theta, 0)$, $B = (0, b \sin \theta)$.

Equation of the line AB (intercept form):

$$\frac{x}{a \cos \theta} + \frac{y}{b \sin \theta} = 1 \Rightarrow \frac{x}{a} \sec \theta + \frac{y}{b} \operatorname{cosec} \theta = 1. \quad (1)$$

Differentiating with respect to θ :

$$\frac{x}{a}(\tan \theta \sec \theta) + \frac{y}{b}(-\cot \theta \operatorname{cosec} \theta) = 0. \quad (2)$$

Multiply (1) by $\cot \theta$ and add (2):

$$\begin{aligned} \frac{x}{a}(\sec \theta \cot \theta + \tan \theta \sec \theta) &= \cot \theta \Rightarrow \frac{x}{a} \sec \theta \left(\frac{\cos \theta}{\sin \theta} + \frac{\sin \theta}{\cos \theta} \right) = \cot \theta \\ &\Rightarrow \frac{x}{a} \sec \theta \left(\frac{\cos^2 \theta + \sin^2 \theta}{\sin \theta \cos \theta} \right) = \cot \theta, \end{aligned}$$

$$\frac{x}{a} \cdot \frac{1}{\cos \theta} \cdot \frac{1}{\sin \theta \cos \theta} = \frac{\cos \theta}{\sin \theta} \Rightarrow \frac{x}{a} = \cos^3 \theta \Rightarrow \cos \theta = \left(\frac{x}{a} \right)^{1/3}.$$

Multiply (1) by $\tan \theta$ and subtract (2):

$$\begin{aligned} \frac{y}{b} \operatorname{cosec} \theta (\tan \theta + \cot \theta) &= \tan \theta \Rightarrow \frac{y}{b} \operatorname{cosec} \theta \left(\frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta} \right) = \frac{\sin \theta}{\cos \theta} \Rightarrow \frac{y}{b} \cdot \frac{1}{\sin \theta} \cdot \frac{1}{\sin \theta} = \sin \theta, \\ \frac{y}{b} &= \sin^3 \theta \Rightarrow \sin \theta = \left(\frac{y}{b} \right)^{1/3}. \end{aligned}$$

Using $\sin^2 \theta + \cos^2 \theta = 1$:

$$\left(\frac{x}{a} \right)^{2/3} + \left(\frac{y}{b} \right)^{2/3} = 1.$$

Problem 6.7. From any point on the circle $x^2 + y^2 = a^2$, perpendiculars are drawn to the axes and the feet of these perpendiculars are joined. Find the envelope of the lines thus formed.

Solution. Parametric form: $x = a \cos \theta$, $y = a \sin \theta$. The line joining $(a \cos \theta, 0)$ and $(0, a \sin \theta)$ is

$$\frac{x}{a \cos \theta} + \frac{y}{a \sin \theta} = 1 \Rightarrow \frac{x}{a} \sec \theta + \frac{y}{a} \operatorname{cosec} \theta = 1. \quad (1)$$

Differentiating with respect to θ :

$$\frac{x}{a}(\sec \theta \tan \theta) + \frac{y}{a}(-\cot \theta \operatorname{cosec} \theta) = 0. \quad (2)$$

Multiply (1) by $\cot \theta$ and add (2):

$$\begin{aligned} \frac{x}{a} \sec \theta (\tan \theta + \cot \theta) &= \cot \theta \Rightarrow \frac{x}{a} \cdot \frac{1}{\cos \theta} \left(\frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta} \right) = \frac{\cos \theta}{\sin \theta} \\ &\Rightarrow \frac{x}{a} = \cos^3 \theta, \quad \cos \theta = \left(\frac{x}{a} \right)^{1/3}. \end{aligned}$$

Multiply (1) by $\tan \theta$ and subtract (2):

$$\begin{aligned} \frac{y}{a} \operatorname{cosec} \theta (\tan \theta + \cot \theta) &= \tan \theta \Rightarrow \frac{y}{a} \operatorname{cosec} \theta \left(\frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta} \right) = \frac{\sin \theta}{\cos \theta} \\ &\Rightarrow \frac{y}{a} = \sin^3 \theta, \quad \sin \theta = \left(\frac{y}{a} \right)^{1/3}. \end{aligned}$$

Squaring and adding,

$$\sin^2 \theta + \cos^2 \theta = \left(\frac{x}{a} \right)^{2/3} + \left(\frac{y}{a} \right)^{2/3} = 1 \Rightarrow \boxed{x^{2/3} + y^{2/3} = a^{2/3}}.$$

The envelope is an astroid.

6.2 Envelope of a two-parameter family of curves

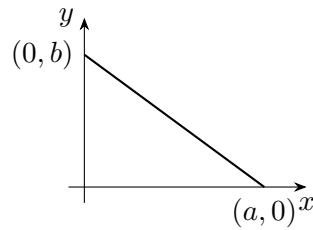
We have a family of curves $f(x, y, a, b) = 0$, where a, b are two parameters subject to the condition $g(a, b) = 0$.

- To get the envelope, differentiate both f and g with respect to any one of the parameters, say a ; we get the relations $\frac{df}{da} = 0$ and $\frac{dg}{da} = 0$ (treating b as a function of a).
- Eliminate a and b using these two relations (together with $f = 0, g = 0$); we get the required envelope.

Problems

Problem 6.8. Find the envelope of the straight lines which slide with their ends on two perpendicular axes such that the sum of the n^{th} powers of the intercepts on the axes is constant.

Solution.



Let a, b be the intercepts on the x - and y -axes respectively. The line is

$$\frac{x}{a} + \frac{y}{b} = 1, \quad (\text{A})$$

and $a^n + b^n = \text{constant}$; let the constant be c^n :

$$a^n + b^n = c^n. \quad (\text{B})$$

Differentiating (A) with respect to a (with b a function of a):

$$-\frac{x}{a^2} - \frac{y}{b^2} \frac{db}{da} = 0 \Rightarrow \frac{db}{da} = -\frac{x}{a^2} \cdot \frac{b^2}{y} = -\left(\frac{b}{a}\right)^2 \frac{x}{y}. \quad (1)$$

Differentiating (B) with respect to a :

$$na^{n-1} + nb^{n-1} \frac{db}{da} = 0 \Rightarrow \frac{db}{da} = -\left(\frac{a}{b}\right)^{n-1}. \quad (2)$$

Equating (1) and (2):

$$\frac{b^2}{a^2} \cdot \frac{x}{y} = \frac{a^{n-1}}{b^{n-1}} \Rightarrow \frac{x}{y} = \frac{a^{n+1}}{b^{n+1}} \Rightarrow \frac{x}{a^{n+1}} = \frac{y}{b^{n+1}} \Rightarrow \frac{x/a}{a^n} = \frac{y/b}{b^n}. \quad (3)$$

By the property of equal ratios (adding numerators and denominators) and using (A), (B):

$$\frac{x/a}{a^n} = \frac{y/b}{b^n} = \frac{\frac{x}{a} + \frac{y}{b}}{a^n + b^n} = \frac{1}{c^n}. \quad (4)$$

From (3) = (4):

$$\frac{x}{a^{n+1}} = \frac{1}{c^n}, \quad \frac{y}{b^{n+1}} = \frac{1}{c^n} \Rightarrow a^{n+1} = xc^n, \quad b^{n+1} = yc^n \Rightarrow a = (xc^n)^{\frac{1}{n+1}}, \quad b = (yc^n)^{\frac{1}{n+1}}.$$

Substituting in (B):

$$(xc^n)^{\frac{n}{n+1}} + (yc^n)^{\frac{n}{n+1}} = c^n \Rightarrow \left(x^{\frac{n}{n+1}} + y^{\frac{n}{n+1}}\right) c^{\frac{n^2}{n+1}} = c^n \Rightarrow x^{\frac{n}{n+1}} + y^{\frac{n}{n+1}} = c^{n - \frac{n^2}{n+1}} = c^{\frac{n}{n+1}}.$$

$$\boxed{x^{n/(n+1)} + y^{n/(n+1)} = c^{n/(n+1)}}$$

Problem 6.9. Find the envelope of the family of ellipses $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where a and b are related by $a^n + b^n = c^n$.

Solution. Differentiating $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with respect to a :

$$x^2(-2a^{-3}) + y^2(-2b^{-3})\frac{db}{da} = 0 \Rightarrow -\frac{2x^2}{a^3} - \frac{2y^2}{b^3}\frac{db}{da} = 0 \Rightarrow \frac{db}{da} = -\frac{b^3}{a^3} \cdot \frac{x^2}{y^2}. \quad (1)$$

Differentiating $a^n + b^n = c^n$ with respect to a :

$$na^{n-1} + nb^{n-1}\frac{db}{da} = 0 \Rightarrow \frac{db}{da} = -\frac{a^{n-1}}{b^{n-1}}. \quad (2)$$

Equating (1) and (2):

$$\frac{b^3}{a^3} \cdot \frac{x^2}{y^2} = \frac{a^{n-1}}{b^{n-1}} \Rightarrow \frac{x^2}{y^2} = \frac{a^{n+2}}{b^{n+2}} \Rightarrow \frac{x^2}{a^{n+2}} = \frac{y^2}{b^{n+2}} \Rightarrow \frac{x^2/a^2}{a^n} = \frac{y^2/b^2}{b^n}.$$

By the property of equal ratios,

$$\frac{x^2/a^2}{a^n} = \frac{y^2/b^2}{b^n} = \frac{\frac{x^2}{a^2} + \frac{y^2}{b^2}}{a^n + b^n} = \frac{1}{c^n}.$$

Hence

$$\frac{x^2}{a^{n+2}} = \frac{1}{c^n}, \quad \frac{y^2}{b^{n+2}} = \frac{1}{c^n} \Rightarrow a = (x^2 c^n)^{\frac{1}{n+2}}, \quad b = (y^2 c^n)^{\frac{1}{n+2}}.$$

Substituting in $a^n + b^n = c^n$:

$$(x^2 c^n)^{\frac{n}{n+2}} + (y^2 c^n)^{\frac{n}{n+2}} = c^n \Rightarrow \left(x^{\frac{2n}{n+2}} + y^{\frac{2n}{n+2}}\right) = \frac{c^n}{(c^n)^{\frac{n}{n+2}}} = (c^n)^{1 - \frac{n}{n+2}} = c^{\frac{2n}{n+2}}.$$

$$\boxed{x^{2n/(n+2)} + y^{2n/(n+2)} = c^{2n/(n+2)}}$$

This is the equation of the required envelope.

7 Relationship between Evolutes and Envelopes

From the definitions of evolute and envelope, we see that

the evolute of a curve $y = f(x)$ is the envelope of the normal lines of the curve.

Problem 7.1. Find the evolute of the parabola $y^2 = 4ax$, treating it as the envelope of its normals.

Solution. We know that the equation of the normal line at (x_1, y_1) for any curve $y = f(x)$ is

$$y - y_1 = -\frac{1}{m}(x - x_1), \quad m = \left. \frac{dy}{dx} \right|_{(x_1, y_1)}. \quad (1)$$

Parametric form of the parabola: $x = at^2$, $y = 2at$, so $x_1 = at^2$, $y_1 = 2at$.

$$\frac{dx}{dt} = 2at, \quad \frac{dy}{dt} = 2a, \quad m = \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{2a}{2at} = \frac{1}{t}.$$

Equation of the normal line:

$$y - 2at = -\frac{1}{1/t}(x - at^2) \Rightarrow y - 2at = -t(x - at^2) = -xt + at^3, \\ y = -xt + at^3 + 2at, \quad (2)$$

which is the equation of the family of normal lines of the parabola, with parameter t .

Differentiating (2) partially with respect to t :

$$0 = -x + a(3t^2) + 2a \Rightarrow 3at^2 = x - 2a \Rightarrow t^2 = \frac{x - 2a}{3a} \quad (\text{an even power of } t).$$

Now (2) gives $y = t(-x + at^2 + 2a)$ (an odd power of t). Squaring,

$$y^2 = t^2 (at^2 + 2a - x)^2.$$

Substituting $t^2 = \frac{x - 2a}{3a}$:

$$y^2 = \left(\frac{x - 2a}{3a} \right) \left(a \left(\frac{x - 2a}{3a} \right) + 2a - x \right)^2 = \left(\frac{x - 2a}{3a} \right) \left(\frac{x - 2a}{3} + 2a - x \right)^2 \\ = \frac{x - 2a}{3a} \left(\frac{x - 2a + 6a - 3x}{3} \right)^2 = \frac{x - 2a}{3a} \left(\frac{-2x + 4a}{3} \right)^2 = \frac{x - 2a}{3a} \cdot \frac{4(x - 2a)^2}{9}.$$

$$\boxed{27ay^2 = 4(x - 2a)^3}$$

is the evolute of the given parabola.

Problem 7.2. Find the evolute of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, treating it as the envelope of its normals.

Solution. Equation of the normal: $y - y_1 = -\frac{1}{m}(x - x_1)$. Parametric equations: $x = a \cos \theta$, $y = b \sin \theta$.

$$\frac{dx}{d\theta} = -a \sin \theta, \quad \frac{dy}{d\theta} = b \cos \theta, \quad m = \frac{dy}{dx} = \frac{b \cos \theta}{-a \sin \theta} = -\frac{b}{a} \cot \theta.$$

Normal at $(a \cos \theta, b \sin \theta)$:

$$y - b \sin \theta = \frac{-1}{-\frac{b}{a} \cot \theta} (x - a \cos \theta) = \frac{a \tan \theta}{b} (x - a \cos \theta) = \frac{a}{b} \cdot \frac{\sin \theta}{\cos \theta} (x - a \cos \theta).$$

Multiplying by $b \cos \theta$: $by \cos \theta - b^2 \sin \theta \cos \theta = ax \sin \theta - a^2 \sin \theta \cos \theta$, i.e.

$$ax \sin \theta - by \cos \theta = (a^2 - b^2) \sin \theta \cos \theta, \quad (1)$$

which is the equation of the family of normal lines of the ellipse, θ being the parameter.

Differentiating (1) partially with respect to θ :

$$\begin{aligned} ax \cos \theta - by(-\sin \theta) &= (a^2 - b^2) (\sin \theta(-\sin \theta) + \cos \theta \cos \theta) \\ \Rightarrow ax \cos \theta + by \sin \theta &= (a^2 - b^2)(\cos^2 \theta - \sin^2 \theta). \end{aligned} \quad (2)$$

Multiply (1) by $\sin \theta$ and (2) by $\cos \theta$, and add:

$$\begin{aligned} \frac{ax \sin^2 \theta - by \sin \theta \cos \theta}{ax \cos^2 \theta + by \sin \theta \cos \theta} &= \frac{(a^2 - b^2) \sin^2 \theta \cos \theta}{(a^2 - b^2)(\cos^2 \theta - \sin^2 \theta) \cos \theta} \\ \frac{ax(\sin^2 \theta + \cos^2 \theta)}{ax(\sin^2 \theta + \cos^2 \theta)} &= \frac{(a^2 - b^2)(\sin^2 \theta \cos \theta + \cos^3 \theta - \sin^2 \theta \cos \theta)}{(a^2 - b^2)(\sin^2 \theta \cos \theta + \cos^3 \theta - \sin^2 \theta \cos \theta)} \\ ax &= (a^2 - b^2) \cos^3 \theta \Rightarrow \left(\frac{ax}{a^2 - b^2} \right)^{1/3} = \cos \theta. \end{aligned} \quad (a)$$

Multiply (1) by $\cos \theta$ and (2) by $\sin \theta$, and subtract:

$$\begin{aligned} \frac{ax \sin \theta \cos \theta - by \cos^2 \theta}{ax \sin \theta \cos \theta + by \sin^2 \theta} &= \frac{(a^2 - b^2) \sin \theta \cos^2 \theta}{(a^2 - b^2)(\cos^2 \theta - \sin^2 \theta) \sin \theta} \\ \frac{-by(\sin^2 \theta + \cos^2 \theta)}{-by(\sin^2 \theta + \cos^2 \theta)} &= \frac{(a^2 - b^2)(\sin \theta \cos^2 \theta - \sin \theta \cos^2 \theta + \sin^3 \theta)}{(a^2 - b^2)(\sin \theta \cos^2 \theta - \sin \theta \cos^2 \theta + \sin^3 \theta)} \\ -by &= (a^2 - b^2) \sin^3 \theta \Rightarrow \left(\frac{by}{a^2 - b^2} \right)^{1/3} = -\sin \theta. \end{aligned} \quad (b)$$

Squaring and adding (a) and (b):

$$\left(\frac{ax}{a^2 - b^2} \right)^{2/3} + \left(\frac{by}{a^2 - b^2} \right)^{2/3} = \cos^2 \theta + \sin^2 \theta = 1 \Rightarrow \frac{(ax)^{2/3} + (by)^{2/3}}{(a^2 - b^2)^{2/3}} = 1,$$

$$\boxed{(ax)^{2/3} + (by)^{2/3} = (a^2 - b^2)^{2/3}}$$

is the evolute of the ellipse.

8 Assignment Problems

Problem 8.1. Show that the measure of curvature of the curve $\sqrt{\frac{x}{a}} + \sqrt{\frac{y}{b}} = 1$ at (x, y) is $\frac{ab}{2(ax + by)^{3/2}}$.

Solution. $\frac{x^{1/2}}{a^{1/2}} + \frac{y^{1/2}}{b^{1/2}} = 1$. Differentiating with respect to x :

$$\frac{1}{\sqrt{a}} \cdot \frac{1}{2} x^{-1/2} + \frac{1}{\sqrt{b}} \cdot \frac{1}{2} y^{-1/2} \frac{dy}{dx} = 0 \Rightarrow \frac{1}{2\sqrt{a}\sqrt{x}} + \frac{1}{2\sqrt{b}\sqrt{y}} \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{1}{2\sqrt{a}\sqrt{x}} \cdot 2\sqrt{b}\sqrt{y},$$

$$y_1 = -\frac{\sqrt{b}}{\sqrt{a}} \cdot \frac{\sqrt{y}}{\sqrt{x}} = -\sqrt{\frac{b}{a}} \left(\frac{y}{x} \right)^{1/2}.$$

Differentiating with respect to x :

$$\begin{aligned} y_2 &= -\sqrt{\frac{b}{a}} \cdot \frac{1}{2} \left(\frac{y}{x}\right)^{-1/2} \left[\frac{x \frac{dy}{dx} - y(1)}{x^2} \right] = -\frac{1}{2} \sqrt{\frac{b}{a}} \left(\frac{x}{y}\right)^{1/2} \frac{1}{x^2} \left(x \cdot \left(-\sqrt{\frac{b}{a}} \frac{\sqrt{y}}{\sqrt{x}} \right) - y \right) \\ &= \frac{1}{2} \sqrt{\frac{b}{a}} \sqrt{\frac{x}{y}} \cdot \frac{1}{x^2} \left(\frac{\sqrt{x}\sqrt{b}\sqrt{y}}{\sqrt{a}} + y \right) = \frac{1}{2} \sqrt{\frac{b}{a}} \sqrt{\frac{x}{y}} \cdot \frac{1}{x^2} \cdot \frac{\sqrt{y}}{\sqrt{a}} (\sqrt{x}\sqrt{b} + \sqrt{y}\sqrt{a}). \end{aligned}$$

From the curve, $\frac{\sqrt{x}}{\sqrt{a}} + \frac{\sqrt{y}}{\sqrt{b}} = 1$, i.e. $\sqrt{x}\sqrt{b} + \sqrt{a}\sqrt{y} = \sqrt{a}\sqrt{b}$. Hence

$$y_2 = \frac{1}{2} \cdot \frac{\sqrt{b}}{\sqrt{a}} \cdot \frac{\sqrt{x}}{\sqrt{y}} \cdot \frac{1}{x^2} \cdot \frac{\sqrt{y}}{\sqrt{a}} \cdot \sqrt{a}\sqrt{b} = \frac{1}{2x^2} \cdot b \frac{\sqrt{x}}{\sqrt{a}} \Rightarrow y_2 = \frac{b\sqrt{x}}{2x^2\sqrt{a}}.$$

Now, with $y_1^2 = \frac{by}{ax}$,

$$\begin{aligned} \rho &= \frac{(1 + y_1^2)^{3/2}}{y_2} = \left(1 + \frac{by}{ax}\right)^{3/2} \cdot \frac{2x^2\sqrt{a}}{b\sqrt{x}} = \frac{2(ax + by)^{3/2}}{(ax)^{3/2} b\sqrt{x}} x^2\sqrt{a} = \frac{2(ax + by)^{3/2}}{b a^{3/2} x^{3/2} x^{1/2}} x^2 a^{1/2}, \\ \rho &= \frac{2(ax + by)^{3/2}}{ab}. \end{aligned}$$

Therefore the curvature is

$$\kappa = \frac{1}{\rho} = \frac{ab}{2(ax + by)^{3/2}}.$$

Problem 8.2. Show that the curves $y = c \cosh\left(\frac{x}{c}\right)$ and $x^2 = 2c(y - c)$ have the same curvature at the points where they cross the y -axis.

Solution. At the y -axis, $x = 0$; both curves pass through $(0, c)$.

Curve 1: $y = c \cosh(x/c)$.

$$\begin{aligned} y_1 &= c \sinh\left(\frac{x}{c}\right) \cdot \frac{1}{c} = \sinh\left(\frac{x}{c}\right), \quad y_1|_{x=0} = 0; \quad y_2 = \cosh\left(\frac{x}{c}\right) \cdot \frac{1}{c}, \quad y_2|_{x=0} = \frac{1}{c}. \\ \rho_1 &= \frac{(1 + y_1^2)^{3/2}}{y_2} = \frac{(1 + 0^2)^{3/2}}{1/c} = c, \quad \kappa_1 = \frac{1}{c}. \end{aligned}$$

Curve 2: $x^2 = 2c(y - c)$. Differentiating, $2x = 2c\left(\frac{dy}{dx} - 0\right)$, so

$$\begin{aligned} y_1 &= \frac{x}{c}, \quad y_1|_{x=0} = 0; \quad y_2 = \frac{1}{c}. \\ \rho_2 &= \frac{(1 + 0)^{3/2}}{1/c} = c, \quad \kappa_2 = \frac{1}{c}. \end{aligned}$$

Since $\kappa_1 = \kappa_2$, both curves have the same curvature at $(0, c)$.

Problem 8.3. Find the equation of the circle of curvature of the curve $\sqrt{x} + \sqrt{y} = \sqrt{a}$ at $\left(\frac{a}{4}, \frac{a}{4}\right)$.

Solution. Differentiating with respect to x :

$$\frac{1}{2} \cdot \frac{1}{\sqrt{x}} + \frac{1}{2} \cdot \frac{1}{\sqrt{y}} \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{1}{2\sqrt{x}} \cdot 2\sqrt{y} \Rightarrow y_1 = -\frac{\sqrt{y}}{\sqrt{x}}; \quad y_1 \Big|_{\left(\frac{a}{4}, \frac{a}{4}\right)} = -1.$$

Differentiating y_1 with respect to x (quotient rule):

$$y_2 = - \left(\frac{\sqrt{x} \cdot \frac{1}{2\sqrt{y}} \frac{dy}{dx} - \sqrt{y} \cdot \frac{1}{2\sqrt{x}}}{(\sqrt{x})^2} \right).$$

Using $\frac{dy}{dx} = -1$ at the given point:

$$y_2 = -\frac{1}{x} \left(-\frac{\sqrt{x}}{2\sqrt{y}} - \frac{\sqrt{y}}{2\sqrt{x}} \right) = \frac{\sqrt{x}}{2x\sqrt{y}} + \frac{\sqrt{y}}{2x\sqrt{x}}.$$

At $\left(\frac{a}{4}, \frac{a}{4}\right)$:

$$y_2 = \frac{\sqrt{a/4}}{2 \cdot \frac{a}{4} \sqrt{a/4}} + \frac{\sqrt{a/4}}{2 \cdot \frac{a}{4} \sqrt{a/4}} = \frac{2}{a} + \frac{2}{a} = \frac{4}{a}.$$

Radius of curvature:

$$\rho = \frac{(1 + y_1^2)^{3/2}}{y_2} = \frac{(1 + 1)^{3/2}}{4/a} = \frac{2^{3/2}a}{4} = \frac{2\sqrt{2}a}{4} = \frac{a}{\sqrt{2}}.$$

Centre of curvature:

$$\bar{x} = x - \frac{y_1}{y_2}(1 + y_1^2) = \frac{a}{4} - \left(\frac{-1}{4/a} \right) (1 + 1^2) = \frac{a}{4} + \frac{a}{4}(2) = \frac{3a}{4},$$

$$\bar{y} = y + \frac{1 + y_1^2}{y_2} = \frac{a}{4} + \frac{1 + 1}{4/a} = \frac{a}{4} + 2 \cdot \frac{a}{4} = \frac{3a}{4}.$$

Circle of curvature $(x - \bar{x})^2 + (y - \bar{y})^2 = \rho^2$:

$$\left(x - \frac{3a}{4}\right)^2 + \left(y - \frac{3a}{4}\right)^2 = \left(\frac{a}{\sqrt{2}}\right)^2 \Rightarrow \frac{(4x - 3a)^2}{4^2} + \frac{(4y - 3a)^2}{4^2} = \frac{a^2}{2},$$

$$\boxed{(4x - 3a)^2 + (4y - 3a)^2 = 8a^2}.$$

Problem 8.4. Prove that the evolute of the curve $x = a \left(\cos \theta + \log \tan \frac{\theta}{2} \right)$, $y = a \sin \theta$ is the catenary $y = a \cosh \left(\frac{x}{a} \right)$.

Solution.

$$\begin{aligned} \frac{dx}{d\theta} &= a \left(-\sin \theta + \frac{1}{\tan \frac{\theta}{2}} \cdot \sec^2 \frac{\theta}{2} \cdot \frac{1}{2} \right) = a \left(-\sin \theta + \frac{\cos \frac{\theta}{2}}{2 \sin \frac{\theta}{2}} \cdot \frac{1}{\cos^2 \frac{\theta}{2}} \right) = a \left(-\sin \theta + \frac{1}{\sin \theta} \right) \\ &= a \left(\frac{-\sin^2 \theta + 1}{\sin \theta} \right) = a \left(\frac{\cos^2 \theta}{\sin \theta} \right), \quad \frac{dy}{d\theta} = a \cos \theta. \end{aligned}$$

$$\frac{dy}{dx} = \frac{a \cos \theta}{a \cos^2 \theta} \cdot \sin \theta \Rightarrow y_1 = \tan \theta.$$

Differentiating with respect to x :

$$y_2 = \sec^2 \theta \cdot \frac{d\theta}{dx} = \sec^2 \theta \cdot \frac{\sin \theta}{a \cos^2 \theta} \Rightarrow y_2 = \frac{1}{a} \sin \theta \sec^4 \theta.$$

$$\begin{aligned} \bar{x} &= x - \frac{y_1}{y_2} (1 + y_1^2) = a \cos \theta + a \log \tan \frac{\theta}{2} - \frac{a \tan \theta}{\sin \theta \sec^4 \theta} (1 + \tan^2 \theta) \\ &= a \cos \theta + a \log \tan \frac{\theta}{2} - \frac{a \sin \theta}{\cos \theta} \cdot \frac{\cos^4 \theta}{\sin \theta} \cdot \frac{1}{\cos^2 \theta} = a \cos \theta + a \log \tan \frac{\theta}{2} - a \cos \theta, \end{aligned}$$

$$\bar{x} = a \log \tan \frac{\theta}{2}.$$

$$\begin{aligned} \bar{y} &= y + \frac{1 + y_1^2}{y_2} = a \sin \theta + \frac{1 + \tan^2 \theta}{\sin \theta \sec^4 \theta} a = a \sin \theta + \frac{a \sec^2 \theta}{\sin \theta \sec^4 \theta} = a \sin \theta + \frac{a}{\sin \theta \sec^2 \theta} \\ &= a \sin \theta + \frac{a \cos^2 \theta}{\sin \theta} = \frac{a \sin^2 \theta + a \cos^2 \theta}{\sin \theta}, \quad \bar{y} = \frac{a}{\sin \theta}. \end{aligned}$$

From $\bar{x} = a \log \tan \frac{\theta}{2}$ we get $e^{\bar{x}/a} = \tan \frac{\theta}{2}$. Using $\sin \theta = \frac{2 \tan \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}}$,

$$\bar{y} = \frac{a (1 + \tan^2 \frac{\theta}{2})}{2 \tan \frac{\theta}{2}} = \frac{a}{2} \cdot \frac{1 + e^{2\bar{x}/a}}{e^{\bar{x}/a}} = \frac{a}{2} \left(\frac{1}{e^{\bar{x}/a}} + \frac{e^{2\bar{x}/a}}{e^{\bar{x}/a}} \right) = \frac{a}{2} (e^{-\bar{x}/a} + e^{\bar{x}/a}).$$

Hence the evolute is $y = a \cosh \left(\frac{x}{a} \right)$.

Problem 8.5. Show that the parabolas $y = -x^2 + x + 1$ and $x = -y^2 + y + 1$ have the same circle of curvature at the point $(1, 1)$.

Solution. Parabola 1: $y = -x^2 + x + 1$.

$$\frac{dy}{dx} = -2x + 1 \Rightarrow y_1|_{(1,1)} = -2 + 1 = -1, \quad y_2 = -2.$$

$$\bar{x} = x - \frac{y_1}{y_2} (1 + y_1^2) = 1 - \frac{-1}{-2} (1 + 1) = 1 - \frac{1}{2} \cdot 2 = 0, \quad \bar{y} = y + \frac{1 + y_1^2}{y_2} = 1 + \frac{1 + 1}{-2} = 1 - 1 = 0.$$

$$\rho = \frac{(1 + y_1^2)^{3/2}}{y_2} = \frac{(1 + 1)^{3/2}}{-2} = -\frac{2\sqrt{2}}{2} = -\sqrt{2} \Rightarrow \rho = \sqrt{2}.$$

Circle of curvature: $(x - 0)^2 + (y - 0)^2 = (\sqrt{2})^2$, i.e. $x^2 + y^2 = 2$.

Parabola 2: $x = -y^2 + y + 1$. Differentiating with respect to x :

$$1 = -2y \frac{dy}{dx} + \frac{dy}{dx} = (1 - 2y) \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{1}{1 - 2y}; \quad y_1|_{(1,1)} = \frac{1}{1 - 2} = -1.$$

$$y_2 = \frac{-1}{(1 - 2y)^2} \left(0 - 2 \frac{dy}{dx} \right) \Rightarrow y_2|_{(1,1)} = \frac{-1}{1} \cdot (2) = -2.$$

$$\bar{x} = 1 - \frac{1}{-2}(1 + 1^2) = 1 - \frac{1}{2} \cdot 2 = 0, \quad \bar{y} = 1 + \frac{1 + 1}{-2} = 1 - 1 = 0,$$

$$\rho = \frac{(1 + 1)^{3/2}}{-2} = -\frac{2\sqrt{2}}{2} = -\sqrt{2} \Rightarrow \rho = \sqrt{2}.$$

Circle of curvature: $x^2 + y^2 = 2$.

Therefore both parabolas have the same circle of curvature $x^2 + y^2 = 2$ at $(1, 1)$.

Problem 8.6. Show that the family of straight lines $2y - 4x + \alpha = 0$ has no envelope, where α is the parameter.

Solution. $f = 2y - 4x + \alpha = 0$. Differentiating partially with respect to α :

$$\frac{\partial f}{\partial \alpha} = 0 - 0 + 1 = 1.$$

Since $\frac{\partial f}{\partial \alpha} = 1 \neq 0$, no relationship can be found between $f = 0$ and $\frac{\partial f}{\partial \alpha} = 0$. Hence the family has no envelope. (The lines are all parallel, with slope 2.)

Problem 8.7. Determine the envelope of the two-parameter family of parabolas $\sqrt{\frac{x}{a}} + \sqrt{\frac{y}{b}} = 1$, where a, b are connected by $a + b = c$, c being a constant.

Solution. $\frac{\sqrt{x}}{\sqrt{a}} + \frac{\sqrt{y}}{\sqrt{b}} = 1$. Differentiating with respect to a :

$$\sqrt{x} \cdot \left(-\frac{1}{2}\right) a^{-3/2} + \sqrt{y} \cdot \left(-\frac{1}{2}\right) b^{-3/2} \frac{db}{da} = 0 \Rightarrow -\frac{\sqrt{x}}{2a^{3/2}} - \frac{\sqrt{y}}{2b^{3/2}} \frac{db}{da} = 0,$$

$$\frac{db}{da} = -\frac{\sqrt{x}}{2a^{3/2}} \cdot \frac{2b^{3/2}}{\sqrt{y}} = -\left(\frac{b}{a}\right)^{3/2} \left(\frac{x}{y}\right)^{1/2}. \quad (1)$$

Differentiating $a + b = c$ with respect to a : $1 + \frac{db}{da} = 0$, so

$$\frac{db}{da} = -1. \quad (2)$$

Equating (1) and (2):

$$\left(\frac{b}{a}\right)^{3/2} \left(\frac{x}{y}\right)^{1/2} = 1 \Rightarrow \frac{x^{1/2}}{a^{3/2}} = \frac{y^{1/2}}{b^{3/2}} \Rightarrow \frac{\sqrt{x}/\sqrt{a}}{a} = \frac{\sqrt{y}/\sqrt{b}}{b}.$$

By the property of equal ratios, and using the given equation,

$$\frac{\sqrt{x}/\sqrt{a}}{a} = \frac{\sqrt{y}/\sqrt{b}}{b} = \frac{\frac{\sqrt{x}}{\sqrt{a}} + \frac{\sqrt{y}}{\sqrt{b}}}{a + b} = \frac{1}{c}.$$

Hence

$$\frac{\sqrt{x}}{a^{3/2}} = \frac{1}{c} \Rightarrow c\sqrt{x} = a^{3/2} \Rightarrow a = (c\sqrt{x})^{2/3}, \quad (3)$$

$$\frac{\sqrt{y}}{b^{3/2}} = \frac{1}{c} \Rightarrow c\sqrt{y} = b^{3/2} \Rightarrow b = (c\sqrt{y})^{2/3}. \quad (4)$$

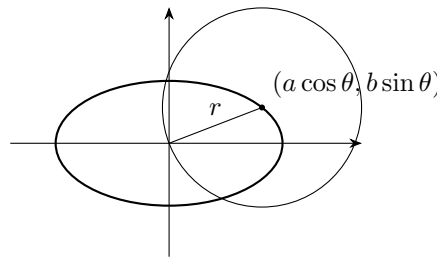
Substituting (3) and (4) in $a + b = c$:

$$(c\sqrt{x})^{2/3} + (c\sqrt{y})^{2/3} = c \Rightarrow c^{2/3} (x^{1/3} + y^{1/3}) = c \Rightarrow x^{1/3} + y^{1/3} = \frac{c}{c^{2/3}},$$

$$\boxed{x^{1/3} + y^{1/3} = c^{1/3}}.$$

Problem 8.8. Find the envelope of the family of circles passing through the origin and having their centres on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Solution.



Any point on the ellipse can be represented as $(a \cos \theta, b \sin \theta)$. A circle with this centre, passing through the origin, has radius r with $r^2 = a^2 \cos^2 \theta + b^2 \sin^2 \theta$. Its equation $(x-x')^2 + (y-y')^2 = r^2$ is

$$\begin{aligned} (x - a \cos \theta)^2 + (y - b \sin \theta)^2 &= (a \cos \theta)^2 + (b \sin \theta)^2, \\ x^2 + a^2 \cos^2 \theta - 2ax \cos \theta + y^2 + b^2 \sin^2 \theta - 2by \sin \theta &= a^2 \cos^2 \theta + b^2 \sin^2 \theta, \\ x^2 + y^2 &= 2ax \cos \theta + 2by \sin \theta, \end{aligned} \quad (1)$$

where θ is the parameter. Differentiating partially with respect to θ :

$$0 = 2ax(-\sin \theta) + 2by \cos \theta. \quad (2)$$

Multiply (1) by $\sin \theta$ and (2) by $\cos \theta$, and add:

$$\begin{aligned} (x^2 + y^2) \sin \theta &= 2ax \sin \theta \cos \theta + 2by \sin^2 \theta \\ 0 \cdot \cos \theta &= -2ax \sin \theta \cos \theta + 2by \cos^2 \theta \\ \hline (x^2 + y^2) \sin \theta &= 2by(\sin^2 \theta + \cos^2 \theta) \end{aligned} \Rightarrow \sin \theta = \frac{2by}{x^2 + y^2}. \quad (3)$$

Multiply (1) by $\cos \theta$ and (2) by $\sin \theta$, and subtract:

$$\begin{aligned} (x^2 + y^2) \cos \theta &= 2ax \cos^2 \theta + 2by \sin \theta \cos \theta \\ 0 \cdot \sin \theta &= -2ax \sin^2 \theta + 2by \sin \theta \cos \theta \\ \hline (x^2 + y^2) \cos \theta &= 2ax(\sin^2 \theta + \cos^2 \theta) \end{aligned} \Rightarrow \cos \theta = \frac{2ax}{x^2 + y^2}. \quad (4)$$

Squaring and adding (3) and (4):

$$\sin^2 \theta + \cos^2 \theta = \left(\frac{2by}{x^2 + y^2} \right)^2 + \left(\frac{2ax}{x^2 + y^2} \right)^2 = 1 \Rightarrow (2by)^2 + (2ax)^2 = (x^2 + y^2)^2,$$

$$\boxed{(x^2 + y^2)^2 = 4a^2x^2 + 4b^2y^2}.$$

Problem 8.9. Determine the evolute of the rectangular hyperbola $\frac{x^2}{a^2} - \frac{y^2}{a^2} = 1$ by treating it as the envelope of its normals.

Solution. Equation of the normal: $(y - y_1) = -\frac{1}{m}(x - x_1)$, $m = \frac{dy}{dx}$.

Parametric form: $x = a \sec \theta$, $y = a \tan \theta$. Then

$$\frac{dx}{d\theta} = a \tan \theta \sec \theta, \quad \frac{dy}{d\theta} = a \sec^2 \theta, \quad m = \frac{dy}{dx} = \frac{a \sec^2 \theta}{a \tan \theta \sec \theta} = \frac{1}{\sin \theta}.$$

Normal at $(a \sec \theta, a \tan \theta)$:

$$y - a \tan \theta = -\frac{1}{1/\sin \theta}(x - a \sec \theta) = -\sin \theta (x - a \sec \theta) \Rightarrow y - a \frac{\sin \theta}{\cos \theta} = -x \sin \theta + a \frac{\sin \theta}{\cos \theta}.$$

Multiplying by $\cos \theta$:

$$y \cos \theta + x \sin \theta \cos \theta = 2a \sin \theta, \quad (1)$$

the equation of the family of normals of the hyperbola (θ is the parameter). Differentiating partially with respect to θ :

$$y(-\sin \theta) + x(\sin \theta(-\sin \theta) + \cos \theta \cos \theta) = 2a \cos \theta \Rightarrow -y \sin \theta + x(\cos^2 \theta - \sin^2 \theta) = 2a \cos \theta. \quad (2)$$

Multiply (1) by $\sin \theta$ and (2) by $\cos \theta$, and add:

$$\begin{array}{rcl} y \sin \theta \cos \theta + x \sin^2 \theta \cos \theta & = & 2a \sin^2 \theta \\ -y \sin \theta \cos \theta + x(\cos^2 \theta - \sin^2 \theta) \cos \theta & = & 2a \cos^2 \theta \\ \hline x \cos \theta (\sin^2 \theta + \cos^2 \theta - \sin^2 \theta) & = & 2a(\sin^2 \theta + \cos^2 \theta) \\ x \cos^3 \theta = 2a & \Rightarrow & \cos \theta = \left(\frac{2a}{x}\right)^{1/3}. \end{array} \quad (3)$$

Multiply (1) by $\cos \theta$ and (2) by $\sin \theta$, and subtract:

$$\begin{array}{rcl} y \cos^2 \theta + x \sin \theta \cos^2 \theta & = & 2a \sin \theta \cos \theta \\ -y \sin^2 \theta + x(\cos^2 \theta - \sin^2 \theta) \sin \theta & = & 2a \sin \theta \cos \theta \\ \hline y(\cos^2 \theta + \sin^2 \theta) + x \sin \theta(\cos^2 \theta - \cos^2 \theta + \sin^2 \theta) & = & 0 \\ y + x \sin^3 \theta = 0 & \Rightarrow & x \sin^3 \theta = -y \Rightarrow -\sin \theta = \left(\frac{y}{x}\right)^{1/3}. \end{array} \quad (4)$$

From (3)² + (4)²:

$$\cos^2 \theta + \sin^2 \theta = \left(\frac{2a}{x}\right)^{2/3} + \left(\frac{y}{x}\right)^{2/3} = 1 \Rightarrow \frac{(2a)^{2/3}}{x^{2/3}} + \frac{y^{2/3}}{x^{2/3}} = 1 \Rightarrow (2a)^{2/3} + y^{2/3} = x^{2/3},$$

$$\boxed{x^{2/3} - y^{2/3} = (2a)^{2/3}}.$$

Problem 8.10. Find the envelope of $\frac{x}{a} + \frac{y}{b} = 1$ subject to the condition $a^m b^n = c^{m+n}$, where a and b are the parameters.

Solution. Differentiating $\frac{x}{a} + \frac{y}{b} = 1$ with respect to a :

$$x(-a^{-2}) + y(-b^{-2}) \frac{db}{da} = 0 \Rightarrow -\frac{x}{a^2} - \frac{y}{b^2} \frac{db}{da} = 0 \Rightarrow \frac{db}{da} = -\frac{b^2}{a^2} \cdot \frac{x}{y}. \quad (1)$$

Differentiating $a^m b^n = c^{m+n}$ with respect to a :

$$a^m \left(n b^{n-1} \frac{db}{da} \right) + b^n (m a^{m-1}) = 0 \Rightarrow \frac{db}{da} = -\frac{m b^n a^{m-1}}{n b^{n-1} a^m} = -\frac{m}{n} \cdot \frac{b}{a}. \quad (2)$$

Equating (1) and (2):

$$-\frac{b^2}{a^2} \cdot \frac{x}{y} = -\frac{m}{n} \cdot \frac{b}{a} \Rightarrow \frac{b}{a} \cdot \frac{x}{y} = \frac{m}{n} \Rightarrow n \frac{x}{a} = m \frac{y}{b} \Rightarrow \frac{x/a}{m} = \frac{y/b}{n}.$$

By the property of equal ratios,

$$\frac{x/a}{m} = \frac{y/b}{n} = \frac{\frac{x}{a} + \frac{y}{b}}{m+n} = \frac{1}{m+n}.$$

Hence²

$$\frac{x}{a} = \frac{m}{m+n} \Rightarrow a = \frac{(m+n)x}{m}, \quad (3)$$

$$\frac{y}{b} = \frac{n}{m+n} \Rightarrow b = \frac{(m+n)y}{n}. \quad (4)$$

Substituting (3) and (4) in $a^m b^n = c^{m+n}$:

$$\left(\frac{(m+n)x}{m} \right)^m \left(\frac{(m+n)y}{n} \right)^n = c^{m+n} \Rightarrow \frac{(m+n)^{m+n}}{m^m n^n} x^m y^n = c^{m+n},$$

$$x^m y^n = \frac{m^m n^n}{(m+n)^{m+n}} c^{m+n}.$$

9 Maxima and Minima of Functions of Two Variables

Suppose we have a function of two variables $f(x, y)$. To find the maxima and minima of such a function:

1. First find $\frac{\partial f}{\partial x} = 0$ and $\frac{\partial f}{\partial y} = 0$ (conditions for *stationary points*). Solving these two equations, we get the values of x and y ; these are the possible maximum or minimum points.
2. Further, find

$$A = \frac{\partial^2 f}{\partial x^2}, \quad B = \frac{\partial^2 f}{\partial x \partial y}, \quad C = \frac{\partial^2 f}{\partial y^2}$$

at each such point (x, y) .

²In the handwritten notes the two ratios appear with m and n interchanged, giving a different final constant. The working here follows directly from $n \frac{x}{a} = m \frac{y}{b}$.

- If $AC - B^2 > 0$ and $A > 0$ (equivalently $C > 0$),³ then (x, y) is a **minimum** point.
- If $AC - B^2 > 0$ and $A < 0$ (equivalently $C < 0$), then (x, y) is a **maximum** point.
- If $AC - B^2 < 0$, the function attains neither a maximum nor a minimum at (x, y) ; such points are called **saddle points**.
- If $AC - B^2 = 0$, further investigation is necessary. It can be carried out by taking points near (x, y) and examining the values of f there.

9.1 Problems

Problem 9.1. Find the extremum (maximum/minimum) of $f = x^3 - y^2 - 3x$.

Solution.

$$\begin{aligned}\frac{\partial f}{\partial x} &= 3x^2 - 3, & \frac{\partial f}{\partial y} &= -2y, \\ A = \frac{\partial^2 f}{\partial x^2} &= 6x, & C = \frac{\partial^2 f}{\partial y^2} &= -2, \\ B = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) &= \frac{\partial(-2y)}{\partial x} = 0.\end{aligned}$$

Solving $\frac{\partial f}{\partial x} = 0$: $3x^2 - 3 = 0 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$. Solving $\frac{\partial f}{\partial y} = 0$: $y = 0$.

Stationary points: $(1, 0)$ and $(-1, 0)$.

	At $(1, 0)$	At $(-1, 0)$
A	6	-6
B	0	0
C	-2	-2
$AC - B^2$	$-12 < 0$	$12 > 0, A < 0$
Nature	neither max nor min	maximum

So f attains a maximum at $(-1, 0)$, and the maximum value is

$$f(-1, 0) = (-1)^3 - 0^2 - 3(-1) = 2.$$

Problem 9.2. Find the extremum of $f = xy(a - x - y)$.

Solution. $f = axy - x^2y - xy^2$.

$$\begin{aligned}\frac{\partial f}{\partial x} &= ay - 2xy - y^2, & \frac{\partial f}{\partial y} &= ax - x^2 - 2xy, \\ A = \frac{\partial^2 f}{\partial x^2} &= -2y, & C = \frac{\partial^2 f}{\partial y^2} &= -2x, & B = \frac{\partial}{\partial x} (ax - x^2 - 2xy) &= a - 2x - 2y. \\ \frac{\partial f}{\partial x} = 0 &\Rightarrow y(a - 2x - y) = 0 \Rightarrow y = 0 \text{ or } a - 2x - y = 0, \\ \frac{\partial f}{\partial y} = 0 &\Rightarrow x(a - x - 2y) = 0 \Rightarrow x = 0 \text{ or } a - x - 2y = 0.\end{aligned}$$

Combining:

³The handwritten notes read “either $A > 0$ or $B > 0$ ”. The standard test uses the sign of A (or, equivalently when $AC - B^2 > 0$, of C), which is what the worked problems below use.

- $x = 0, y = 0 \Rightarrow (0, 0)$;
- $x = 0, a - 2x - y = 0 \Rightarrow (0, a)$;
- $y = 0, a - x - 2y = 0 \Rightarrow (a, 0)$;
- $a - 2x - y = 0, a - x - 2y = 0 \Rightarrow 2x + y = a, x + 2y = a \Rightarrow x = y = \frac{a}{3}$, i.e. $\left(\frac{a}{3}, \frac{a}{3}\right)$.

At $(0, 0)$: $A = 0, B = a, C = 0$; $AC - B^2 = -a^2 < 0 \Rightarrow$ neither max nor min.

At $(0, a)$: $A = -2a, B = a - 0 - 2a = -a, C = 0$; $AC - B^2 = -a^2 < 0 \Rightarrow$ neither max nor min.

At $(a, 0)$: $A = 0, B = a - 2a - 0 = -a, C = -2a$; $AC - B^2 = -a^2 < 0 \Rightarrow$ neither max nor min.

At $\left(\frac{a}{3}, \frac{a}{3}\right)$:

$$A = -\frac{2a}{3}, \quad B = a - \frac{2a}{3} - \frac{2a}{3} = -\frac{a}{3}, \quad C = -\frac{2a}{3}, \quad AC - B^2 = \frac{4a^2}{9} - \frac{a^2}{9} = \frac{3a^2}{9} = \frac{a^2}{3} > 0.$$

Hence $\left(\frac{a}{3}, \frac{a}{3}\right)$ is a *maximum* point if $a > 0$ (then $A < 0$) and a *minimum* point if $a < 0$ (then $A > 0$). The extreme value is $f\left(\frac{a}{3}, \frac{a}{3}\right) = \frac{a^3}{27}$.

Problem 9.3. Find the extremum of the function $f = 2 + 2x + 2y - x^2 - y^2$.

Solution.

$$\frac{\partial f}{\partial x} = 2 - 2x, \quad \frac{\partial f}{\partial y} = 2 - 2y, \quad \frac{\partial^2 f}{\partial x^2} = -2, \quad \frac{\partial^2 f}{\partial y^2} = -2, \quad \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial(2 - 2y)}{\partial x} = 0.$$

$$\frac{\partial f}{\partial x} = 0 \Rightarrow x = 1; \quad \frac{\partial f}{\partial y} = 0 \Rightarrow y = 1. \quad \text{At } (1, 1): A = -2, B = 0, C = -2,$$

$$AC - B^2 = (-2)(-2) - 0 = 4 > 0, \quad A < 0.$$

Therefore $(1, 1)$ is a maximum point, and the maximum value of f is

$$f(1, 1) = 2 + 2(1) + 2(1) - 1^2 - 1^2 = 4.$$

10 Constrained Maxima and Minima (Lagrange Multipliers)

- Suppose we have to find the extremum of $f(x, y, z)$ subject to the condition $\phi(x, y, z) = 0$.
- For such cases, construct a new function $g(x, y, z)$ such that

$$g = f + \lambda\phi,$$

where λ is called the *Lagrange multiplier*.

- For this new function, consider $\frac{\partial g}{\partial x} = 0, \frac{\partial g}{\partial y} = 0, \frac{\partial g}{\partial z} = 0$. Solving these equations (together with $\phi = 0$), we get the value of λ as well as the point (x, y, z) , which represents the relative maximum or minimum point.

Note. The point obtained by this method is a relative maximum/minimum point, and we can conclude whether it is an actual maximum or minimum from the geometrical interpretation of the problem.

10.1 Problems

Problem 10.1. Find the rectangle of perimeter l which has maximum area.

Solution. Let x, y be the length and breadth of the rectangle. Given perimeter l : $2x + 2y = l$. Area $A = xy$.

Here we have to find the rectangle of maximum area for the given perimeter. Hence

$$f = xy, \quad \phi = 2x + 2y - l, \quad g = f + \lambda\phi = xy + \lambda(2x + 2y - l).$$

$$\frac{\partial g}{\partial x} = y + 2\lambda = 0 \Rightarrow y = -2\lambda, \quad \frac{\partial g}{\partial y} = x + 2\lambda = 0 \Rightarrow x = -2\lambda.$$

Therefore $x = y$: the square is the only rectangle which gives the maximum area for the given perimeter.

Problem 10.2. A rectangular box, open at the top, is to have a volume of 32 cm^3 . Find the dimensions of the box which require the least material for its construction.

Solution. Let x, y, z be the length, breadth and height. The least material means the least surface area:

$$f = 2(x + y)z + xy = 2xz + 2yz + xy, \quad \phi = xyz - 32.$$

$$g = xy + 2yz + 2zx + \lambda(xyz - 32).$$

$$\frac{\partial g}{\partial x} = y + 2z + \lambda yz = 0, \tag{1}$$

$$\frac{\partial g}{\partial y} = x + 2z + \lambda xz = 0, \tag{2}$$

$$\frac{\partial g}{\partial z} = 2y + 2x + \lambda xy = 0. \tag{3}$$

Dividing (1) by yz , (2) by xz and (3) by xy :

$$\frac{1}{z} + \frac{2}{y} = -\lambda \quad (4), \quad \frac{1}{z} + \frac{2}{x} = -\lambda \quad (5), \quad \frac{2}{x} + \frac{2}{y} = -\lambda \quad (6).$$

$$(4) - (5): 2\left(\frac{1}{y} - \frac{1}{x}\right) = 0 \Rightarrow x = y.$$

$$\text{Then (6) gives } \frac{2}{y} + \frac{2}{y} = -\lambda \Rightarrow \frac{4}{y} = -\lambda \Rightarrow y = -\frac{4}{\lambda}, \text{ and } x = -\frac{4}{\lambda}.$$

$$\text{From (5): } \frac{1}{z} = -\lambda - \frac{2}{x} = -\lambda - \frac{2\lambda}{-4} = -\lambda + \frac{\lambda}{2} = -\frac{\lambda}{2} \Rightarrow z = -\frac{2}{\lambda}.$$

We know $xyz = 32$:

$$\left(-\frac{4}{\lambda}\right)\left(-\frac{4}{\lambda}\right)\left(-\frac{2}{\lambda}\right) = 32 \Rightarrow -\frac{32}{\lambda^3} = 32 \Rightarrow \lambda = -1.$$

Hence $x = 4$, $y = 4$, $z = 2$: length = 4 cm, breadth = 4 cm, height = 2 cm are the required dimensions.

Problem 10.3. Find the point on the plane $2x + 3y - z = 5$ which is nearest to the origin.

Solution. The distance between the origin $(0, 0, 0)$ and a point (x, y, z) must be a minimum. By the distance formula,

$$d = \sqrt{(x-0)^2 + (y-0)^2 + (z-0)^2} \Rightarrow d^2 = x^2 + y^2 + z^2.$$

$$f = x^2 + y^2 + z^2, \quad \phi = 2x + 3y - z - 5, \quad g = x^2 + y^2 + z^2 + \lambda(2x + 3y - z - 5).$$

$$\frac{\partial g}{\partial x} = 2x + 2\lambda = 0 \Rightarrow x = -\lambda, \quad \frac{\partial g}{\partial y} = 2y + 3\lambda = 0 \Rightarrow y = -\frac{3\lambda}{2}, \quad \frac{\partial g}{\partial z} = 2z - \lambda = 0 \Rightarrow z = \frac{\lambda}{2}.$$

Substituting in $2x + 3y - z = 5$:

$$2(-\lambda) + 3\left(-\frac{3\lambda}{2}\right) - \frac{\lambda}{2} = 5 \Rightarrow \frac{-4\lambda - 9\lambda - \lambda}{2} = 5 \Rightarrow -14\lambda = 10 \Rightarrow \lambda = -\frac{5}{7}.$$

$$x = \frac{5}{7}, \quad y = \frac{3}{2} \cdot \frac{5}{7} = \frac{15}{14}, \quad z = -\frac{5}{7} \cdot \frac{1}{2} = -\frac{5}{14}.$$

The required point is $\left(\frac{5}{7}, \frac{15}{14}, -\frac{5}{14}\right)$.

Homework

Problem 10.4. Find the minimum distance from the point $(3, 4, 15)$ to the cone $x^2 + y^2 = 4z^2$.

Solution. Let (x, y, z) be a point on the cone; the given point is $(3, 4, 15)$. We minimise

$$f = d^2 = (x-3)^2 + (y-4)^2 + (z-15)^2, \quad \phi = x^2 + y^2 - 4z^2 = 0,$$

$$g = (x-3)^2 + (y-4)^2 + (z-15)^2 + \lambda(x^2 + y^2 - 4z^2).$$

$$\frac{\partial g}{\partial x} = 2(x-3) + 2\lambda x = 0 \Rightarrow 2x(\lambda+1) - 6 = 0 \Rightarrow x = \frac{3}{\lambda+1},$$

$$\frac{\partial g}{\partial y} = 2(y-4) + 2\lambda y = 0 \Rightarrow 2y(\lambda+1) - 8 = 0 \Rightarrow y = \frac{4}{\lambda+1},$$

$$\frac{\partial g}{\partial z} = 2(z-15) - 8\lambda z = 0 \Rightarrow 2z(1-4\lambda) - 30 = 0 \Rightarrow z = \frac{15}{1-4\lambda}.$$

Substituting in $x^2 + y^2 = 4z^2$:

$$\frac{9}{(\lambda+1)^2} + \frac{16}{(\lambda+1)^2} = \frac{4 \cdot 225}{(1-4\lambda)^2} \Rightarrow \frac{25}{(\lambda+1)^2} = \frac{900}{(1-4\lambda)^2}.$$

Taking the square root,

$$\frac{5}{\lambda+1} = \frac{30}{1-4\lambda} \Rightarrow 5 - 20\lambda = 30\lambda + 30 \Rightarrow -25 = 50\lambda \Rightarrow \lambda = -\frac{1}{2}.$$

Then $x = \frac{3}{1/2} = 6$, $y = \frac{4}{1/2} = 8$, $z = \frac{15}{1+2} = 5$. The nearest point is $(6, 8, 5)$, and

$$d = \sqrt{(6-3)^2 + (8-4)^2 + (5-15)^2} = \sqrt{9 + 16 + 100} = \sqrt{125} = 5\sqrt{5} \text{ units.}$$

Problem 10.5. Find the shortest and the longest distance from the point $(1, 2, -1)$ to the sphere $x^2 + y^2 + z^2 = 24$.

Solution.

$$f = d^2 = (x - 1)^2 + (y - 2)^2 + (z + 1)^2, \quad \phi = x^2 + y^2 + z^2 - 24,$$

$$g = (x - 1)^2 + (y - 2)^2 + (z + 1)^2 + \lambda(x^2 + y^2 + z^2 - 24).$$

$$\frac{\partial g}{\partial x} = 2(x - 1) + 2\lambda x = 0 \Rightarrow 2x(\lambda + 1) - 2 = 0 \Rightarrow x = \frac{1}{\lambda + 1},$$

$$\frac{\partial g}{\partial y} = 2(y - 2) + 2\lambda y = 0 \Rightarrow 2y(\lambda + 1) - 4 = 0 \Rightarrow y = \frac{2}{\lambda + 1},$$

$$\frac{\partial g}{\partial z} = 2(z + 1) + 2\lambda z = 0 \Rightarrow 2z(\lambda + 1) + 2 = 0 \Rightarrow z = -\frac{1}{\lambda + 1}.$$

So $\lambda + 1 = \frac{1}{x} = \frac{2}{y} = -\frac{1}{z}$, giving $y = 2x$ and $z = -x$. Substituting in $x^2 + y^2 + z^2 = 24$:

$$x^2 + 4x^2 + x^2 = 24 \Rightarrow 6x^2 = 24 \Rightarrow x = \pm 2, \quad y = \pm 4, \quad z = \mp 2.$$

The points are $P_1(2, 4, -2)$ and $P_2(-2, -4, 2)$.

$$d_1 = \sqrt{(2 - 1)^2 + (4 - 2)^2 + (-2 + 1)^2} = \sqrt{1 + 4 + 1} = \sqrt{6} \approx 2.45 \text{ units (shortest distance),}$$

$$d_2 = \sqrt{(-2 - 1)^2 + (-4 - 2)^2 + (2 + 1)^2} = \sqrt{9 + 36 + 9} = \sqrt{54} \approx 7.35 \text{ units (longest distance).}$$

Problem 10.6. Divide 120 into three parts so that the sum of their products taken two at a time is maximum.

Solution. Let the parts be x, y, z with $x + y + z = 120$; we maximise $xy + yz + zx$.

$$f = xy + yz + zx, \quad \phi = x + y + z - 120, \quad g = xy + yz + zx + \lambda(x + y + z - 120).$$

$$\frac{\partial g}{\partial x} = y + z + \lambda = 0, \quad \frac{\partial g}{\partial y} = x + z + \lambda = 0, \quad \frac{\partial g}{\partial z} = y + x + \lambda = 0.$$

Subtracting the first two: $y - x = 0 \Rightarrow y = x$. Then $y + x + \lambda = 0 \Rightarrow 2x + \lambda = 0 \Rightarrow x = -\frac{\lambda}{2}$,
 $y = -\frac{\lambda}{2}$. From $x + z = -\lambda$: $z = -\lambda + \frac{\lambda}{2} = -\frac{\lambda}{2}$.

Now $x + y + z = 120$ gives $-\frac{3\lambda}{2} = 120 \Rightarrow \lambda = -80$. Hence

$$x = y = z = -\frac{(-80)}{2} = 40.$$

The three parts are 40, 40, 40.

Problem 10.7. Find the maximum value of $x^m y^n z^p$ when $x + y + z = a$.

Solution.

$$f = x^m y^n z^p, \quad \phi = x + y + z - a, \quad g = x^m y^n z^p + \lambda(x + y + z - a).$$

$$\frac{\partial g}{\partial x} = m x^{m-1} y^n z^p + \lambda = 0, \quad \frac{\partial g}{\partial y} = n x^m y^{n-1} z^p + \lambda = 0, \quad \frac{\partial g}{\partial z} = p x^m y^n z^{p-1} + \lambda = 0.$$

Dividing each by $x^m y^n z^p$:

$$\frac{m}{x} = \frac{-\lambda}{x^m y^n z^p}, \quad \frac{n}{y} = \frac{-\lambda}{x^m y^n z^p}, \quad \frac{p}{z} = \frac{-\lambda}{x^m y^n z^p},$$

so that

$$\frac{m}{x} = \frac{n}{y} = \frac{p}{z}.$$

By the property of equal ratios,

$$\frac{m}{x} = \frac{n}{y} = \frac{p}{z} = \frac{m+n+p}{x+y+z} = \frac{m+n+p}{a}.$$

Therefore

$$x = \frac{am}{m+n+p}, \quad y = \frac{an}{m+n+p}, \quad z = \frac{ap}{m+n+p},$$

and the maximum value is

$$x^m y^n z^p = \frac{a^{m+n+p} m^m n^n p^p}{(m+n+p)^{m+n+p}}.$$

Homework problems

1. Divide 24 into three parts such that the continued product of the first, the square of the second and the cube of the third is maximum. ($f = xy^2 z^3$, $\phi = x + y + z - 24$)
2. Show that the rectangular solid of maximum volume that can be inscribed in a sphere is a cube. ($f = xyz$, $\phi = x^2 + y^2 + z^2 - a^2$)
3. Find the area of the greatest rectangle that can be inscribed in an ellipse. ($f = xy$, $\phi = \frac{x^2}{a^2} + \frac{y^2}{b^2} - 1$)
4. The pressure P at any point (x, y, z) in space is $P = 400xyz^2$. Find the highest pressure on the surface of the unit sphere $x^2 + y^2 + z^2 = 1$.
5. A wire of length b is cut into two parts which are bent in the form of a square and a circle respectively. Find the least value of the sum of the areas so found.

Solutions to the homework problems

Problem 10.8. Divide 24 into three parts such that the continued product of the first, the square of the second and the cube of the third is maximum.

Solution. $f = xyz^3$, $\phi = x + y + z - 24$, $g = xyz^3 + \lambda(x + y + z - 24)$.

$$\frac{\partial g}{\partial x} = y^2 z^3 + \lambda = 0 \Rightarrow \lambda = -y^2 z^3,$$

$$\frac{\partial g}{\partial y} = 2xyz^3 + \lambda = 0 \Rightarrow \lambda = -2xyz^3,$$

$$\frac{\partial g}{\partial z} = 3xy^2 z^2 + \lambda = 0 \Rightarrow \lambda = -3xy^2 z^2.$$

Hence $y^2 z^3 = 2xyz^3 = 3xy^2 z^2$.

- $y^2 z^3 = 2xyz^3 \Rightarrow y = 2x$.
- $y^2 z^3 = 3xy^2 z^2 \Rightarrow z = 3x$.

Then $x + y + z = 24 \Rightarrow x + 2x + 3x = 24 \Rightarrow 6x = 24$, so

$$x = 4, \quad y = 8, \quad z = 12.$$

Problem 10.9. Show that the rectangular solid of maximum volume that can be inscribed in a sphere is a cube.

Solution. Let x, y, z be the length, breadth and height of the rectangular solid.

$$f = xyz, \quad \phi = x^2 + y^2 + z^2 - a^2, \quad g = xyz + \lambda(x^2 + y^2 + z^2 - a^2).$$

$$\frac{\partial g}{\partial x} = yz + 2\lambda x = 0 \Rightarrow \lambda = -\frac{yz}{2x},$$

$$\frac{\partial g}{\partial y} = xz + 2\lambda y = 0 \Rightarrow \lambda = -\frac{xz}{2y},$$

$$\frac{\partial g}{\partial z} = xy + 2\lambda z = 0 \Rightarrow \lambda = -\frac{xy}{2z}.$$

Therefore

$$\frac{yz}{2x} = \frac{xz}{2y} = \frac{xy}{2z}.$$

From the first equality, $y^2 = x^2 \Rightarrow y = x$ (1). From the second, $z^2 = y^2 \Rightarrow z = y$ (2).

From (1) and (2), $x = y = z$: length = breadth = height. Hence the rectangular solid is a cube.

Problem 10.10. Find the area of the greatest rectangle that can be inscribed in the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Solution.

$$f = xy \text{ (area measure of the rectangle)}, \quad \phi = \frac{x^2}{a^2} + \frac{y^2}{b^2} - 1, \quad g = xy + \lambda \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} - 1 \right).$$

$$\frac{\partial g}{\partial x} = y + \frac{2x}{a^2} \lambda = 0 \Rightarrow \lambda = -\frac{ya^2}{2x}, \quad \frac{\partial g}{\partial y} = x + \frac{2y}{b^2} \lambda = 0 \Rightarrow \lambda = -\frac{xb^2}{2y}.$$

$$\text{Equating, } \frac{ya^2}{2x} = \frac{xb^2}{2y} \Rightarrow y^2 a^2 = x^2 b^2 \Rightarrow ay = bx \Rightarrow x = \frac{ay}{b}.$$

Substituting in the ellipse:

$$\frac{a^2}{b^2} \cdot \frac{y^2}{a^2} + \frac{y^2}{b^2} = 1 \Rightarrow \frac{2y^2}{b^2} = 1 \Rightarrow y = \frac{b}{\sqrt{2}}, \quad x = \frac{a}{b} \cdot \frac{b}{\sqrt{2}} = \frac{a}{\sqrt{2}}.$$

Hence

$$xy = \frac{a}{\sqrt{2}} \cdot \frac{b}{\sqrt{2}} = \frac{ab}{2}.$$

Remark. Here $f = xy$ is taken as the area measure, as in the notes. If (x, y) is the corner of the rectangle in the first quadrant, the full rectangle has sides $2x$ and $2y$, and its area is $4xy = 2ab$.

Problem 10.11. The pressure at any point (x, y, z) in space is $P = 400xyz^2$. Find the highest pressure on the surface of the unit sphere $x^2 + y^2 + z^2 = 1$.

Solution.

$$f = 400xyz^2, \quad \phi = x^2 + y^2 + z^2 - 1, \quad g = 400xyz^2 + \lambda(x^2 + y^2 + z^2 - 1).$$

$$\begin{aligned} \frac{\partial g}{\partial x} &= 400yz^2 + 2\lambda x = 0 \Rightarrow \lambda = -\frac{200yz^2}{x}, \\ \frac{\partial g}{\partial y} &= 400xz^2 + 2\lambda y = 0 \Rightarrow \lambda = -\frac{200xz^2}{y}, \\ \frac{\partial g}{\partial z} &= 400x \cdot 2zy + 2\lambda z = 800xyz + 2\lambda z = 0 \Rightarrow \lambda = -400xy. \end{aligned}$$

So

$$\frac{200yz^2}{x} = \frac{200xz^2}{y} = 400xy.$$

From the first equality, $y^2 = x^2 \Rightarrow y = x$. From the second, $\frac{200xz^2}{y} = 400xy \Rightarrow z^2 = 2y^2 \Rightarrow z = \sqrt{2}y$.

Substituting in $x^2 + y^2 + z^2 = 1$: $y^2 + y^2 + 2y^2 = 1 \Rightarrow 4y^2 = 1$, so

$$y = \frac{1}{2}, \quad x = \frac{1}{2}, \quad z = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}.$$

Therefore the highest pressure is

$$P = 400 \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = 50.$$

Problem 10.12. A wire of length b is cut into two parts which are bent in the form of a square and a circle respectively. Find the least value of the sum of the areas so found.

Solution. Let the two parts be x and y , so $x + y = b$.

Square (perimeter x): side = $\frac{x}{4}$, area = $\frac{x^2}{16}$. *Circle* (circumference y): $2\pi r = y \Rightarrow r = \frac{y}{2\pi}$,
area = $\pi r^2 = \frac{y^2}{4\pi}$.

$$f = \frac{x^2}{16} + \frac{y^2}{4\pi}, \quad \phi = x + y - b, \quad g = \frac{x^2}{16} + \frac{y^2}{4\pi} + \lambda(x + y - b).$$

$$\frac{\partial g}{\partial x} = \frac{2x}{16} + \lambda = \frac{x}{8} + \lambda = 0 \Rightarrow x = -8\lambda, \quad \frac{\partial g}{\partial y} = \frac{2y}{4\pi} + \lambda = \frac{y}{2\pi} + \lambda = 0 \Rightarrow y = -2\pi\lambda.$$

Since $x + y = b$: $-8\lambda - 2\pi\lambda = b \Rightarrow \lambda = -\frac{b}{8 + 2\pi}$. Hence

$$x = \frac{8b}{8 + 2\pi} = \frac{4b}{4 + \pi}, \quad y = \frac{2\pi b}{8 + 2\pi} = \frac{\pi b}{4 + \pi}.$$

The least total area is

$$f = \left(\frac{4b}{4 + \pi}\right)^2 \frac{1}{16} + \left(\frac{\pi b}{4 + \pi}\right)^2 \frac{1}{4\pi} = \frac{b^2}{(4 + \pi)^2} + \frac{\pi b^2}{4(4 + \pi)^2} = \frac{4b^2 + \pi b^2}{4(4 + \pi)^2} = \frac{b^2(4 + \pi)}{4(4 + \pi)^2},$$

$$A = \frac{b^2}{4(4 + \pi)} \approx 0.035 b^2.^4$$

Problem 10.13. Find the volume of the greatest rectangular parallelepiped inscribed in the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.

Solution. Let the dimensions of the parallelepiped be $2x, 2y, 2z$, so the volume is $V = 2x \cdot 2y \cdot 2z$.

$$f = 8xyz, \quad \phi = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1, \quad g = 8xyz + \lambda \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 \right).$$

$$\frac{\partial g}{\partial x} = 8yz + \frac{2x}{a^2}\lambda = 0, \quad \frac{\partial g}{\partial y} = 8xz + \frac{2y}{b^2}\lambda = 0, \quad \frac{\partial g}{\partial z} = 8xy + \frac{2z}{c^2}\lambda = 0,$$

$$\lambda = -\frac{4yza^2}{x} = -\frac{4xzb^2}{y} = -\frac{4xyc^2}{z}.$$

From the first equality, $\frac{yza^2}{x} = \frac{xzb^2}{y} \Rightarrow \frac{y^2}{b^2} = \frac{x^2}{a^2}$; from the second, $\frac{z^2}{c^2} = \frac{y^2}{b^2}$. Hence

$$\frac{x^2}{a^2} = \frac{y^2}{b^2} = \frac{z^2}{c^2} = k.$$

Since $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$, we get $3k = 1 \Rightarrow k = \frac{1}{3}$. So

$$x = \frac{a}{\sqrt{3}}, \quad y = \frac{b}{\sqrt{3}}, \quad z = \frac{c}{\sqrt{3}},$$

$$V = 8xyz = 8 \cdot \frac{a}{\sqrt{3}} \cdot \frac{b}{\sqrt{3}} \cdot \frac{c}{\sqrt{3}} = \frac{8abc}{3\sqrt{3}}.$$

⁴In the first working in the notes the factor 4 in the denominator was dropped (giving $b^2/(4 + \pi) \approx 0.14b^2$); the class-test solution of the same problem later in the notes gives the correct value $b^2/(4(4 + \pi))$.

Problem 10.14. A rectangular box open at the top is to have a volume of 32 cc. Find the dimensions of the box that require the least material for its construction. (Alternative elimination for Problem 10.2.)

Solution. As before, $f = xy + 2yz + 2xz$, $\phi = xyz - 32$, and

$$y + 2z = -\lambda yz, \quad x + 2z = -\lambda xz, \quad 2y + 2x = -\lambda xy.$$

Dividing by yz , xz , xy respectively:

$$\frac{1}{z} + \frac{2}{y} = -\lambda \quad (1), \quad \frac{1}{z} + \frac{2}{x} = -\lambda \quad (2), \quad \frac{2}{x} + \frac{2}{y} = -\lambda \quad (3).$$

$$(1) - (2): 2\left(\frac{1}{y} - \frac{1}{x}\right) = 0 \Rightarrow x = y. \quad (2) - (3): \frac{1}{z} - \frac{2}{y} = 0 \Rightarrow \frac{1}{z} = \frac{2}{y} \Rightarrow y = 2z.$$

Now $xyz = 32 \Rightarrow y \cdot y \cdot \frac{y}{2} = 32 \Rightarrow y^3 = 64$, so⁵

$$y = 4, \quad x = 4, \quad z = 2.$$

11 Class Test I

1. A flat circular plate is heated so that the temperature at any point (x, y) is $u(x, y) = x^2 + 2y^2 - x$. Find the coldest point on the plate.
2. A manufacturing company finds that its profit from assembling a certain number of bicycles is given by the formula $f(x, y) = x^3 + y^3 - 12x - 3y + 20$. Find the maximum profit point.
3. Suppose a closed rectangular box has length twice its breadth and has constant volume V . Determine the dimensions of the box requiring least surface area.
4. A wire of length b is cut into two parts, which are bent in the form of a square and a circle respectively. Find the least value of the sum of the areas so found.
5. Prove that the rectangular solid of maximum volume which can be inscribed in a sphere is a cube.

Solutions

Problem 11.1. Coldest point of $u = x^2 + 2y^2 - x$.

Solution.

$$\frac{\partial u}{\partial x} = 2x + 0 - 1, \quad \frac{\partial u}{\partial y} = 0 + 4y - 0,$$

$$A = \frac{\partial^2 u}{\partial x^2} = 2, \quad C = \frac{\partial^2 u}{\partial y^2} = 4,$$

$$B = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial y} \right) = \frac{\partial(4y)}{\partial x} = 0.$$

$AC - B^2 = (2)(4) - 0^2 = 8 > 0$ and $A > 0$, so the stationary point is a minimum point.

⁵The handwritten notes record $y = 8$ at this step; since $y^3 = 64$ gives $y = 4$, the values here agree with the first solution of this problem.

$$\frac{\partial u}{\partial x} = 0 \Rightarrow 2x - 1 = 0 \Rightarrow x = \frac{1}{2}; \quad \frac{\partial u}{\partial y} = 0 \Rightarrow 4y = 0 \Rightarrow y = 0.$$

Coldest point: $\left(\frac{1}{2}, 0\right)$.

Problem 11.2. Maximum profit point of $f = x^3 + y^3 - 12x - 3y + 20$.^a

^aIn the handwritten notes the term appears as $+3y$; with $+3y$, $\partial f/\partial y = 3y^2 + 3$ never vanishes. The working in the notes ($y = \pm 1$) corresponds to $-3y$, which is used here.

Solution.

$$\frac{\partial f}{\partial x} = 3x^2 - 12, \quad \frac{\partial f}{\partial y} = 3y^2 - 3, \quad \frac{\partial^2 f}{\partial x^2} = 6x, \quad \frac{\partial^2 f}{\partial y^2} = 6y, \quad \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial(3y^2 - 3)}{\partial x} = 0.$$

$$\frac{\partial f}{\partial x} = 0 \Rightarrow 3x^2 = 12 \Rightarrow x^2 = 4 \Rightarrow x = \pm 2; \quad \frac{\partial f}{\partial y} = 0 \Rightarrow 3y^2 = 3 \Rightarrow y = \pm 1.$$

Stationary points: $(2, 1)$, $(-2, 1)$, $(2, -1)$, $(-2, -1)$.

Point	$A = 6x$	B	$C = 6y$	$AC - B^2$
$(2, 1)$	12	0	6	$72 > 0, A > 0 \Rightarrow$ minimum
$(-2, 1)$	-12	0	6	$-72 < 0 \Rightarrow$ neither
$(2, -1)$	12	0	-6	$-72 < 0 \Rightarrow$ neither
$(-2, -1)$	-12	0	-6	$72 > 0, A < 0 \Rightarrow$ maximum

So f is maximum at $(-2, -1)$:

$$f(-2, -1) = (-2)^3 + (-1)^3 - 12(-2) - 3(-1) + 20 = -8 - 1 + 24 + 3 + 20 = 38.$$

Maximum profit is attained at the point $(-2, -1)$.

Problem 11.3. Closed box, length twice the breadth, volume V , least surface area.

Solution. Let breadth = b , length = $2b$, height = h . Volume: $lbh = (2b)(b)(h) = 2b^2h = V$.

Surface area:

$$2(l + b)h + 2lb = 2lh + 2bh + 2lb = 2(2b)h + 2bh + 2(2b)(b) = 4bh + 2bh + 4b^2,$$

$$f = 4b^2 + 6bh, \quad \phi = 2b^2h - V, \quad g = 4b^2 + 6bh + \lambda(2b^2h - V).$$

$$\frac{\partial g}{\partial b} = 8b + 6h + \lambda(4bh) = 0 \Rightarrow \lambda = -\frac{8b + 6h}{4bh} = -\frac{4b + 3h}{2bh}, \quad (1)$$

$$\frac{\partial g}{\partial h} = 0 + 6b + \lambda(2b^2) = 0 \Rightarrow \lambda = -\frac{6b}{2b^2} = -\frac{3}{b}. \quad (2)$$

Equating (1) and (2):

$$\frac{4b + 3h}{2bh} = \frac{3}{b} \Rightarrow 4b + 3h = 6h \Rightarrow 4b = 3h \Rightarrow h = \frac{4}{3}b.$$

Using $\phi = 0$:

$$V = 2b^2h = 2b^2 \cdot \frac{4b}{3} = \frac{8b^3}{3} \Rightarrow b^3 = \frac{3V}{8} \Rightarrow \boxed{b = \frac{(3V)^{1/3}}{2}}.$$

$$h = \frac{4}{3} \cdot \frac{(3V)^{1/3}}{2} \Rightarrow \boxed{h = \frac{2}{3}(3V)^{1/3}}, \quad l = 2b \Rightarrow \boxed{l = (3V)^{1/3}}.$$

Problem 11.4. Wire of length b cut into a square and a circle; least total area.

Solution. Let $x + y = b$, where x forms the square and y the circle.

Square: $4a = x \Rightarrow a = \frac{x}{4}$. Circle: circumference $2\pi r = y \Rightarrow r = \frac{y}{2\pi}$.

$$\text{Area} = a^2 + \pi r^2 = \left(\frac{x}{4}\right)^2 + \pi \left(\frac{y}{2\pi}\right)^2 = \frac{x^2}{16} + \frac{y^2}{4\pi}.$$

$$f = \frac{x^2}{16} + \frac{y^2}{4\pi}, \quad \phi = x + y - b, \quad g = f + \lambda \phi.$$

$$\frac{\partial g}{\partial x} = \frac{2x}{16} + \lambda = \frac{x}{8} + \lambda = 0 \Rightarrow x = -8\lambda, \quad \frac{\partial g}{\partial y} = \frac{2y}{4\pi} + \lambda = \frac{y}{2\pi} + \lambda = 0 \Rightarrow y = -2\pi\lambda.$$

$$x + y = b \Rightarrow -8\lambda - 2\pi\lambda = b \Rightarrow \lambda = -\frac{b}{8 + 2\pi}. \text{ Hence}$$

$$x = \frac{8b}{8 + 2\pi} = \frac{4b}{4 + \pi}, \quad y = \frac{2\pi b}{8 + 2\pi} = \frac{\pi b}{4 + \pi}.$$

$$\text{Area} = \frac{(4b)^2}{(4 + \pi)^2} \cdot \frac{1}{16} + \frac{(\pi b)^2}{(4 + \pi)^2} \cdot \frac{1}{4\pi} = \frac{b^2}{(4 + \pi)^2} + \frac{\pi b^2}{4(4 + \pi)^2} = \frac{4b^2 + \pi b^2}{4(4 + \pi)^2} = \frac{b^2(4 + \pi)}{4(4 + \pi)^2},$$

$$\boxed{A = \frac{b^2}{4(4 + \pi)}} \Rightarrow \text{least area.}$$

Problem 11.5. Rectangular solid of maximum volume inscribed in a sphere.

Solution. Let x, y, z be the length, breadth and height of the rectangular solid.

$$f = xyz, \quad \phi = x^2 + y^2 + z^2 - a^2, \quad g = xyz + \lambda(x^2 + y^2 + z^2 - a^2).$$

$$\frac{\partial g}{\partial x} = yz + 2\lambda x = 0, \quad \frac{\partial g}{\partial y} = xz + 2\lambda y = 0, \quad \frac{\partial g}{\partial z} = xy + 2\lambda z = 0,$$

$$\lambda = -\frac{yz}{2x} = -\frac{xz}{2y} = -\frac{xy}{2z}.$$

$$\text{From } \frac{yz}{2x} = \frac{xz}{2y}: y^2 = x^2 \Rightarrow y = x \quad (1). \text{ From } \frac{xz}{2y} = \frac{xy}{2z}: z^2 = y^2 \Rightarrow z = y \quad (2).$$

From (1) and (2), $x = y = z$, i.e. length = breadth = height. The rectangular solid is a cube.

12 Additional Problems

Problem 12.1. Show that the line joining any point t on the cycloid $x = a(t + \sin t)$, $y = a(1 - \cos t)$ to its centre of curvature is bisected by the line $y = 2a$.

Solution.

$$\frac{dx}{dt} = a(1 + \cos t), \quad \frac{dy}{dt} = a(0 + \sin t) = a \sin t.$$

Using $1 + \cos t = 2 \cos^2 \frac{t}{2}$,

$$\frac{dy}{dx} = \frac{a \sin t}{a(1 + \cos t)} = \frac{2 \sin \frac{t}{2} \cos \frac{t}{2}}{2 \cos^2 \frac{t}{2}} \Rightarrow y_1 = \tan \frac{t}{2}.$$

$$y_2 = \sec^2 \frac{t}{2} \cdot \frac{1}{2} \cdot \frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{t}{2} \cdot \frac{1}{a(1 + \cos t)} = \frac{1}{2} \sec^2 \frac{t}{2} \cdot \frac{\sec^2 \frac{t}{2}}{2a} \Rightarrow y_2 = \frac{\sec^4 \frac{t}{2}}{4a}.$$

The ordinate of the centre of curvature is

$$\begin{aligned} \bar{y} &= y + \frac{1 + y_1^2}{y_2} = a(1 - \cos t) + \frac{1 + \tan^2 \frac{t}{2}}{\sec^4 \frac{t}{2}} \cdot 4a = a(1 - \cos t) + \frac{\sec^2 \frac{t}{2}}{\sec^4 \frac{t}{2}} \cdot 4a \\ &= a(1 - \cos t) + \frac{4a}{\sec^2 \frac{t}{2}} = a(1 - \cos t) + 2a \left(2 \cos^2 \frac{t}{2} \right) = a(1 - \cos t) + 2a(1 + \cos t) \\ &= a - a \cos t + 2a + 2a \cos t = 3a + a \cos t. \end{aligned}$$

The ordinate of the mid-point of the line joining (x, y) and $(\bar{x}, \bar{y})$ is

$$\frac{y + \bar{y}}{2} = \frac{a(1 - \cos t) + 3a + a \cos t}{2} = \frac{a - a \cos t + 3a + a \cos t}{2} = \frac{4a}{2} = 2a.$$

Hence the mid-point lies on the line $y = 2a$; that is, the line joining the point to its centre of curvature is bisected by $y = 2a$.

Problem 12.2. Find the envelope of the family of straight lines which, together with the line segments intercepted by the coordinate axes, form triangles of equal area.

Solution. Let the line be

$$\frac{x}{a} + \frac{y}{b} = 1. \quad (1)$$

The area of the triangle formed with the axes is $\frac{1}{2}ab = \text{constant}$, so

$$ab = k \quad (\text{constant}). \quad (2)$$

Differentiating (1) with respect to a :

$$-\frac{x}{a^2} - \frac{y}{b^2} \frac{db}{da} = 0 \Rightarrow \frac{db}{da} = -\frac{x}{a^2} \cdot \frac{b^2}{y}.$$

Differentiating (2) with respect to a :

$$a \frac{db}{da} + b = 0 \Rightarrow \frac{db}{da} = -\frac{b}{a}.$$

Equating,

$$\frac{b^2}{a^2} \cdot \frac{x}{y} = \frac{b}{a} \Rightarrow \frac{b}{a} \cdot \frac{x}{y} = 1 \Rightarrow \frac{x}{a} = \frac{y}{b} = \Lambda \text{ (say).}$$

From (1), $\Lambda + \Lambda = 1 \Rightarrow \Lambda = \frac{1}{2}$. Hence

$$\frac{x}{a} = \frac{1}{2} \Rightarrow a = 2x, \quad \frac{y}{b} = \frac{1}{2} \Rightarrow b = 2y.$$

Substituting in (2): $ab = k \Rightarrow (2x)(2y) = k$, so the envelope is

$$\boxed{4xy = k},$$

a rectangular hyperbola.